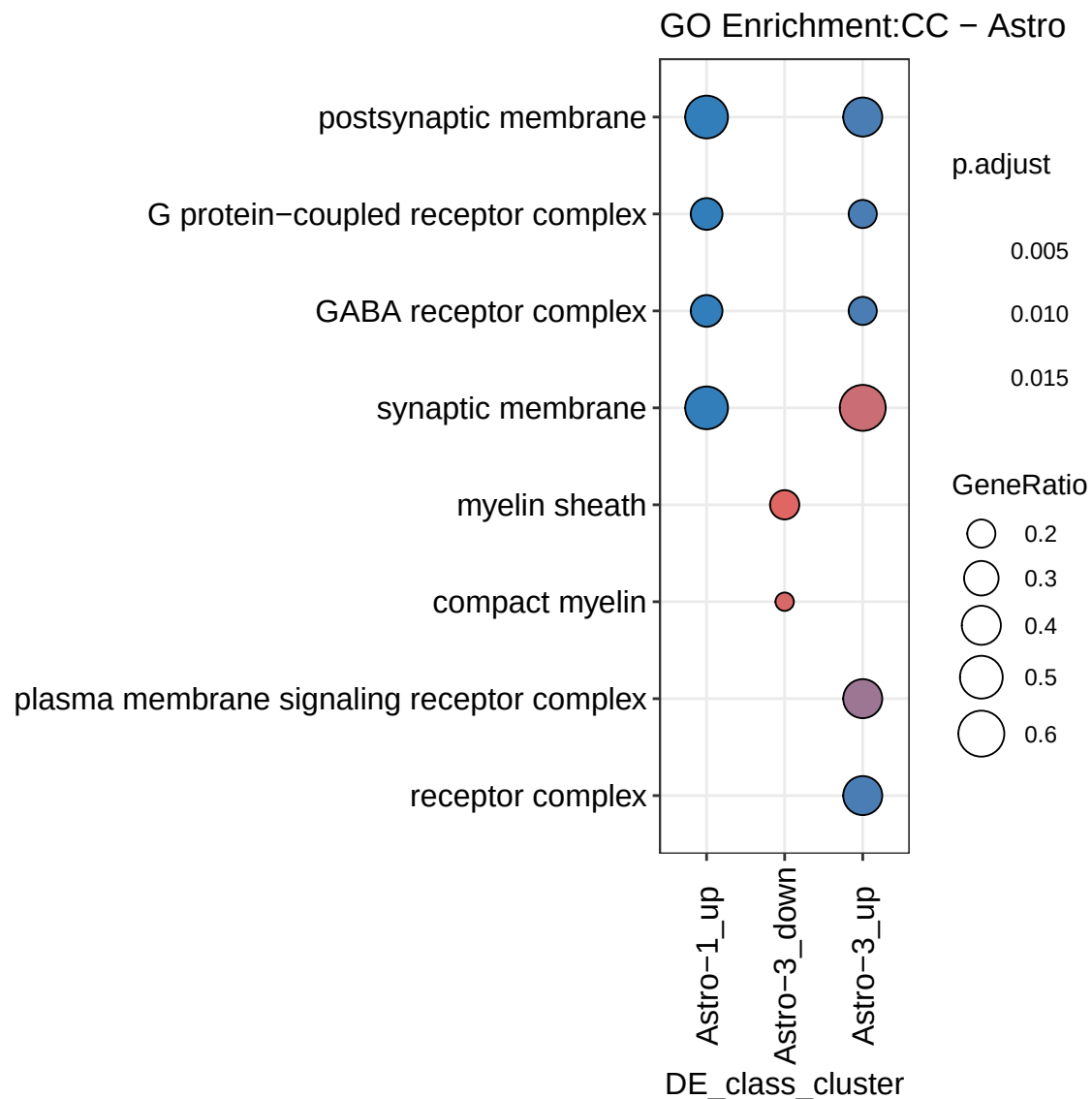
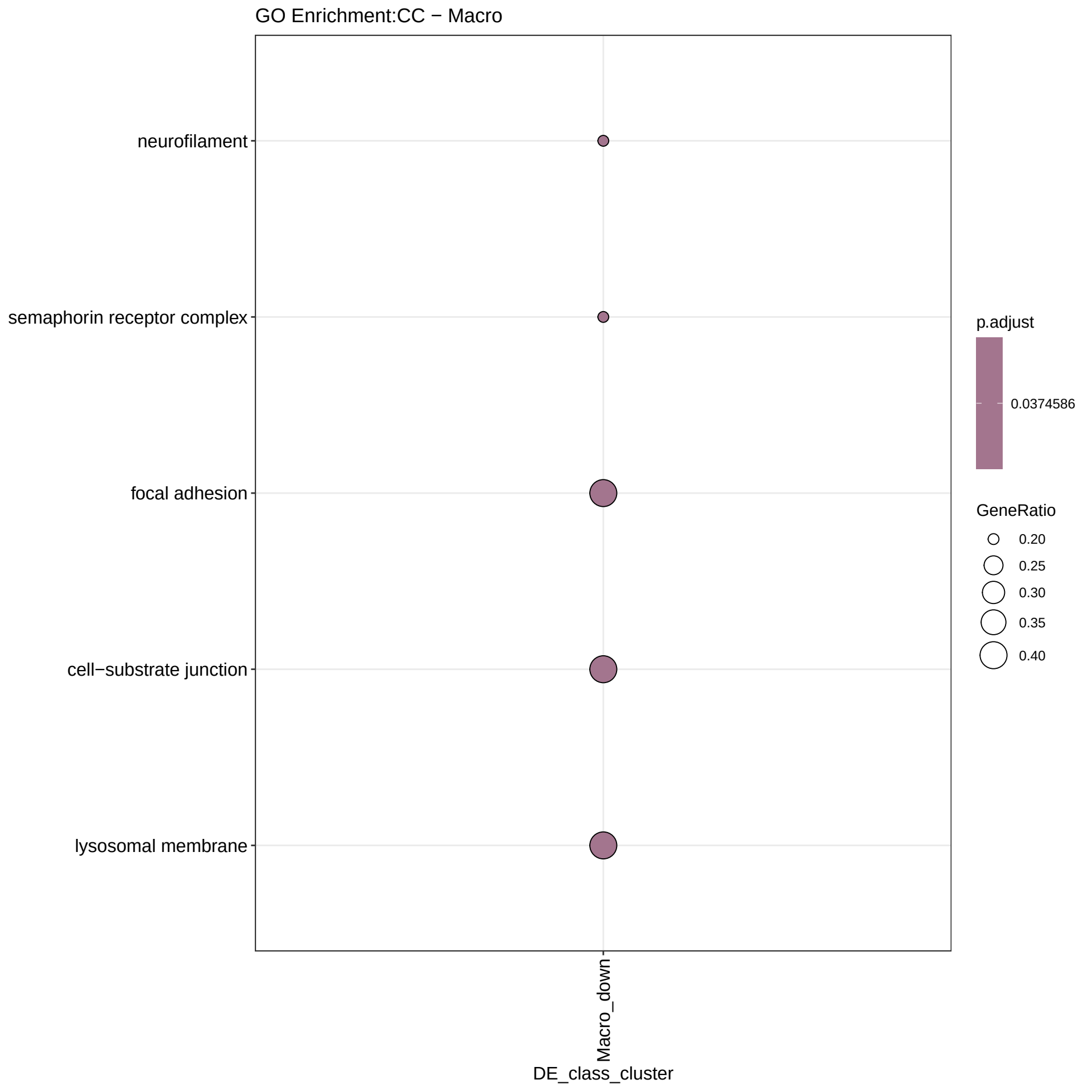
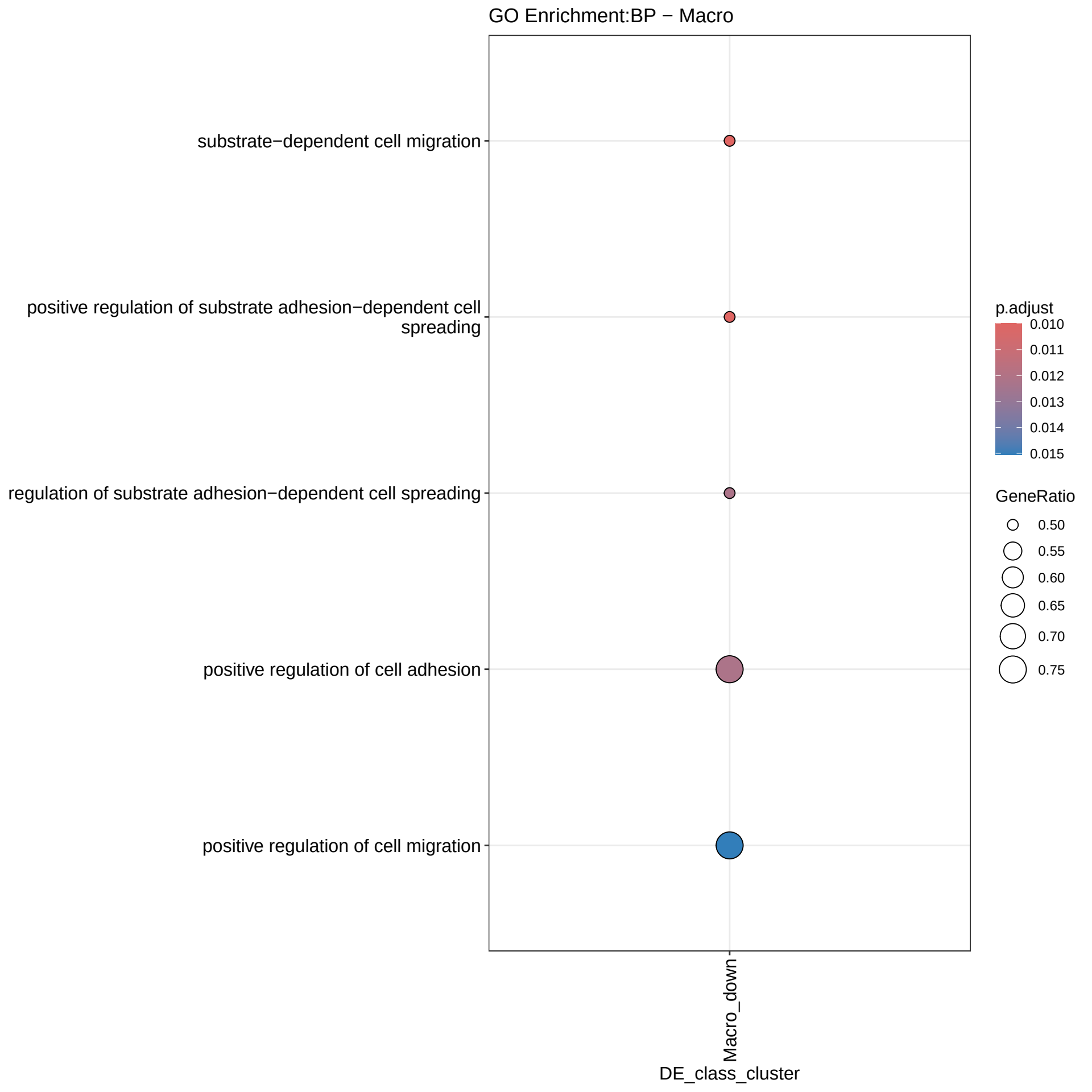
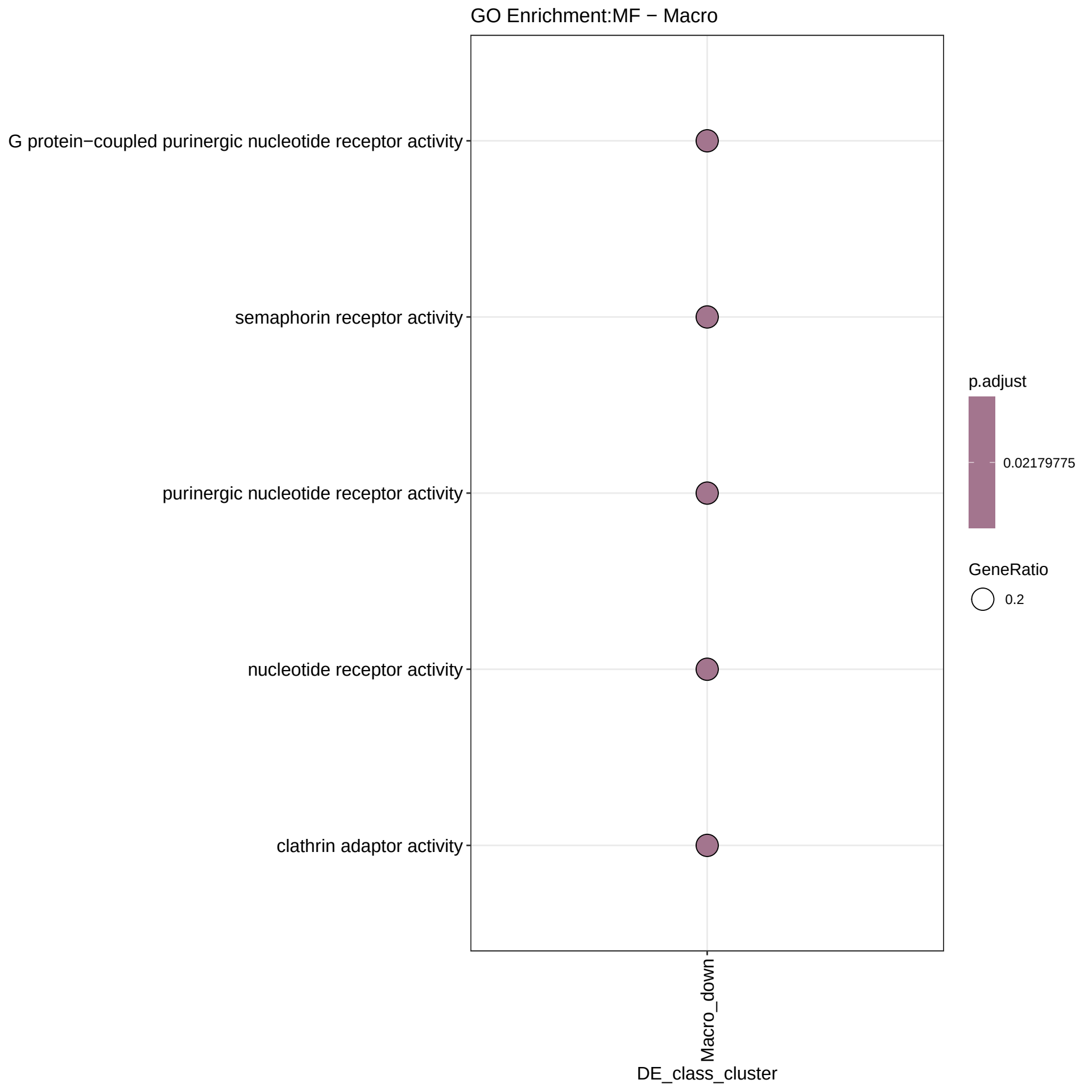
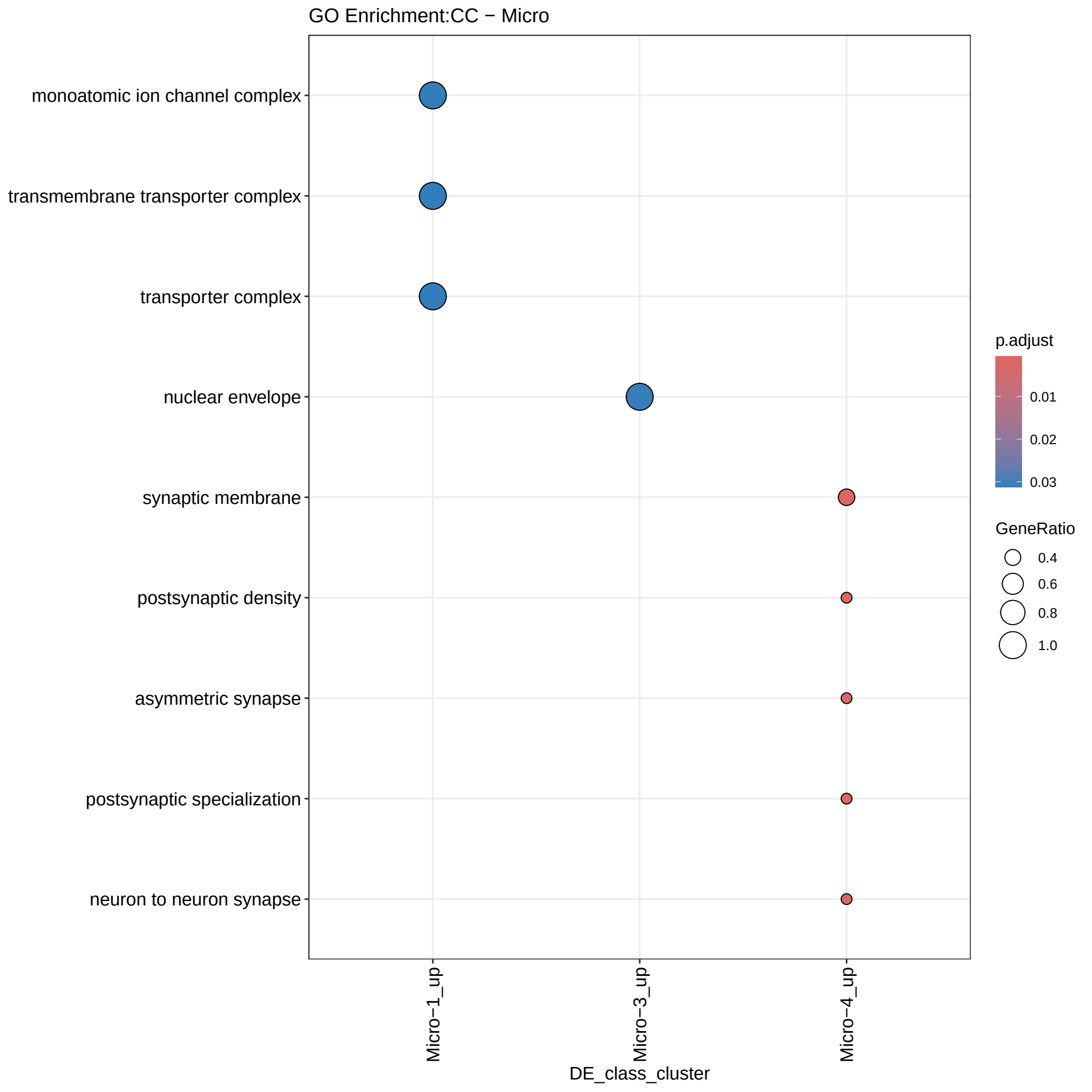
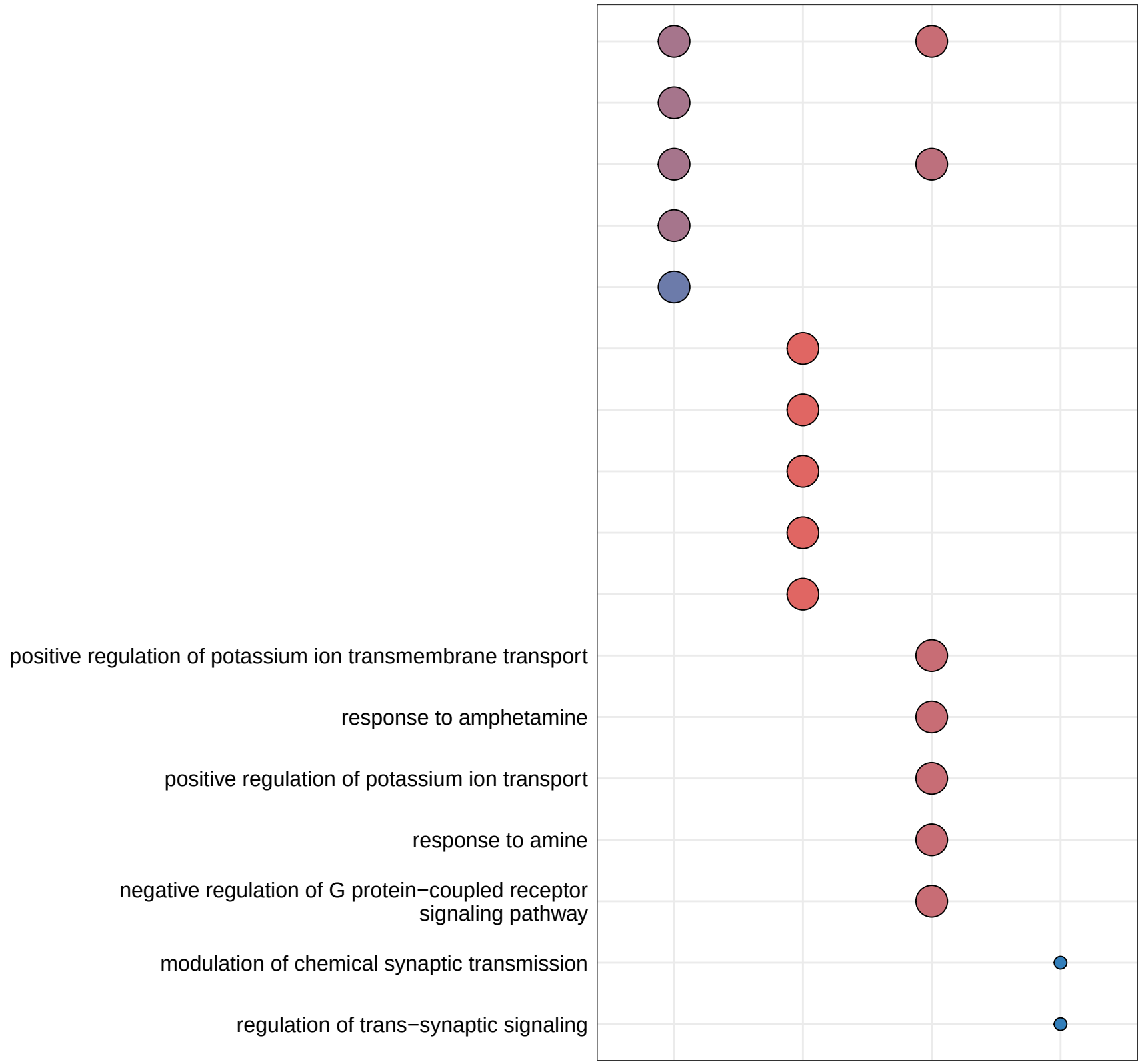

GO Enrichment:CC − Astro
postsynaptic membrane
G protein−coupled receptor complex
GABA receptor complex
synaptic membrane
myelin sheath
compact myelin
plasma membrane signaling receptor complex
receptor complex
Astro−1_up
Astro−3_down
Astro−3_up
DE_class_cluster
p.adjust
0.005
0.010
0.015
GeneRatio
0.2
0.3
0.4
0.5
0.6

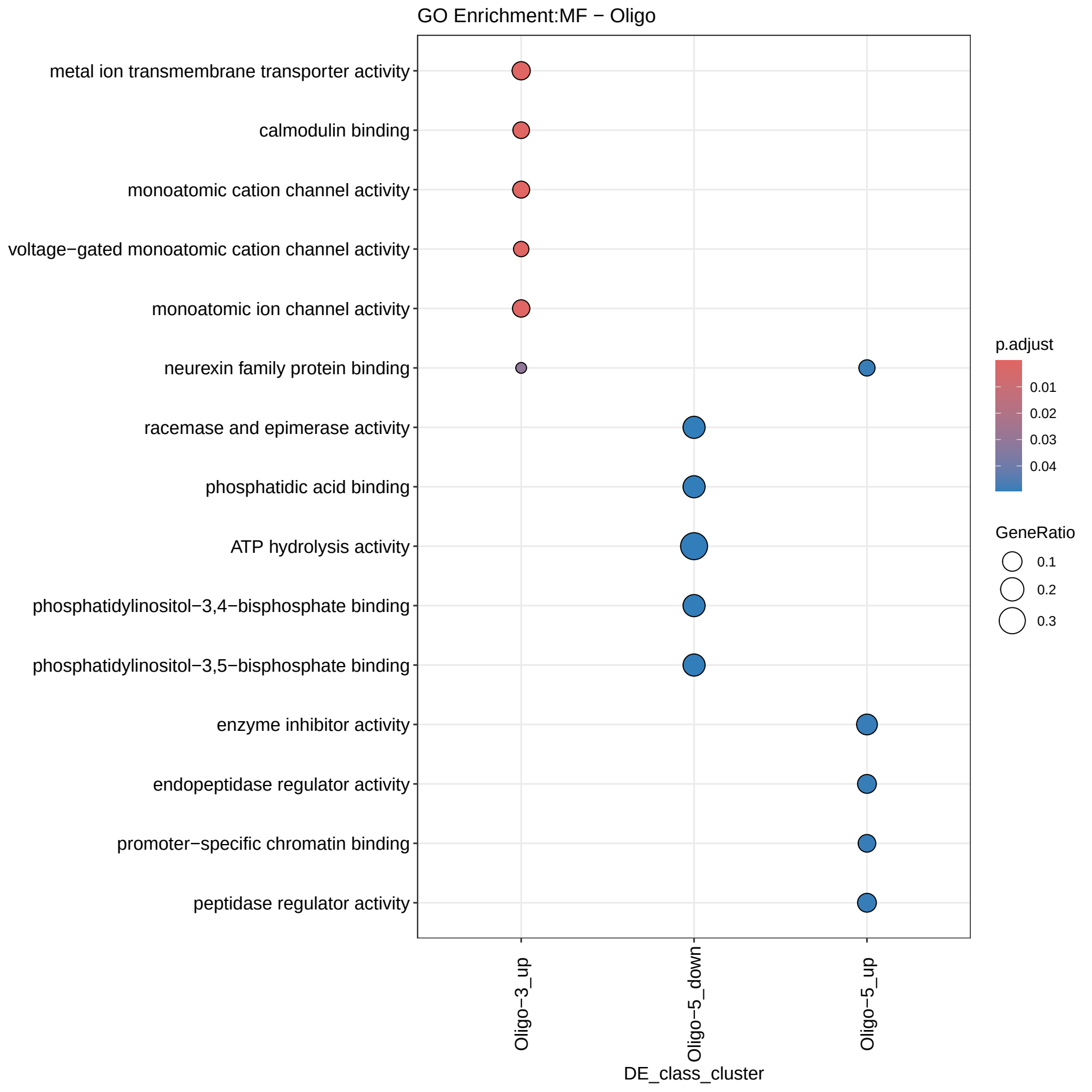

GO Enrichment:CC − Macro
neurofilament
semaphorin receptor complex
focal adhesion
cell−substrate junction
lysosomal membrane
Macro_down
DE_class_cluster
p.adjust
0.0374586
GeneRatio
0.20
0.25
0.30
0.35
0.40

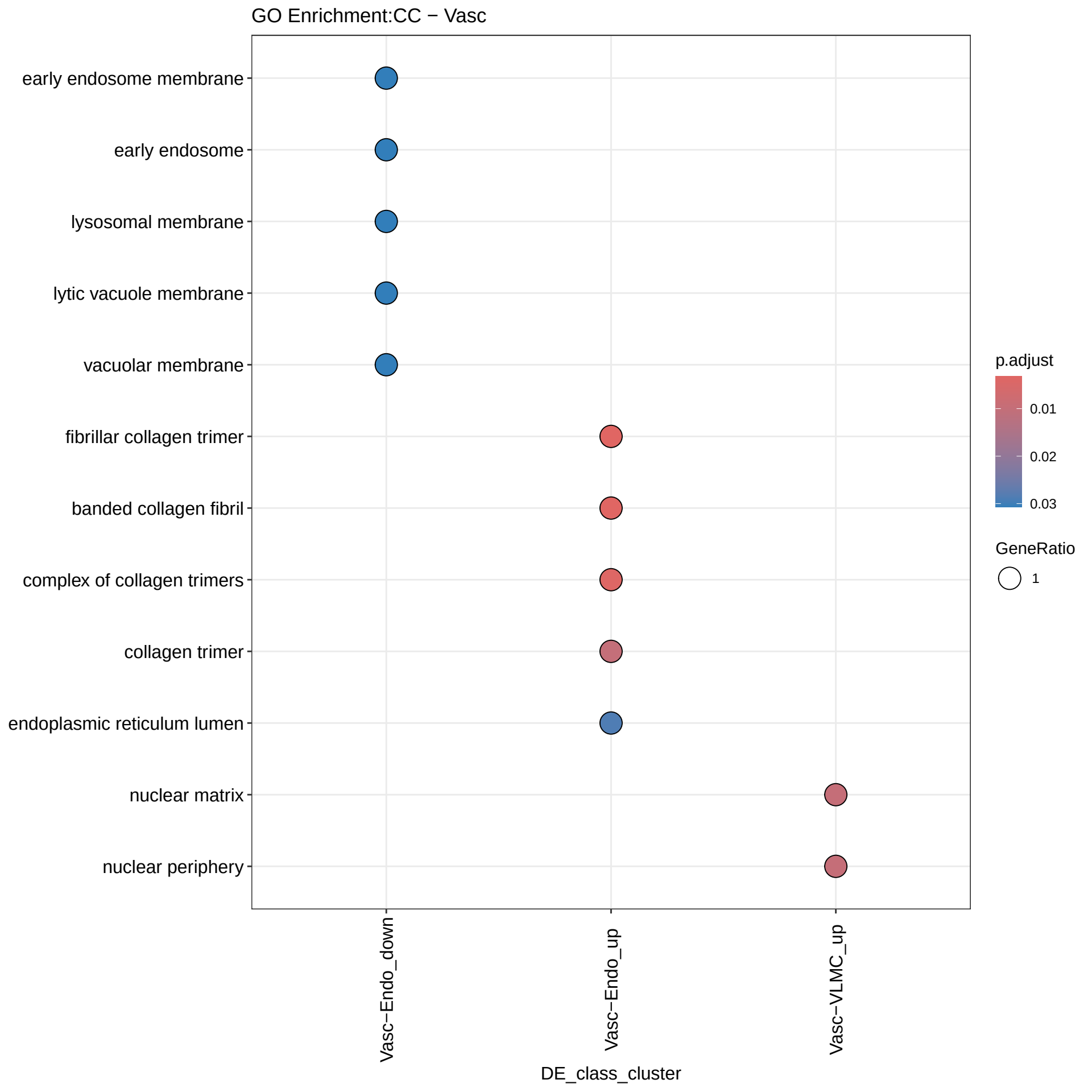

GO Enrichment:BP − Macro
substrate−dependent cell migration
positive regulation of substrate adhesion−dependent cell spreading
regulation of substrate adhesion−dependent cell spreading
positive regulation of cell adhesion
positive regulation of cell migration
Macro_down
DE_class_cluster
p.adjust
0.010
0.011
0.012
0.013
0.014
0.015
GeneRatio
0.50
0.55
0.60
0.65
0.70
0.75

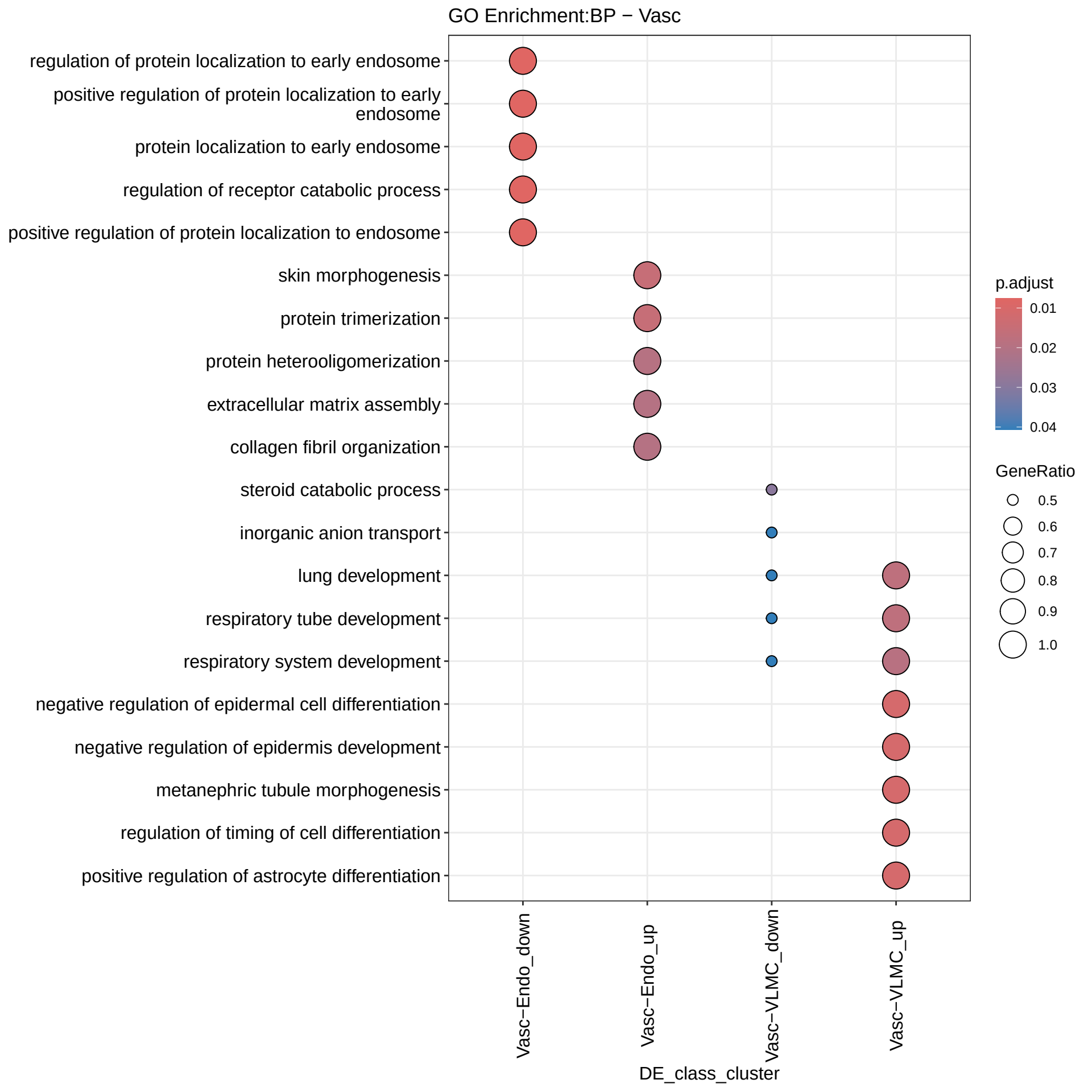

GO Enrichment:MF − Macro
G protein−coupled purinergic nucleotide receptor activity
semaphorin receptor activity
purinergic nucleotide receptor activity
nucleotide receptor activity
clathrin adaptor activity
Macro_down
DE_class_cluster
p.adjust
0.02179775
GeneRatio
0.2

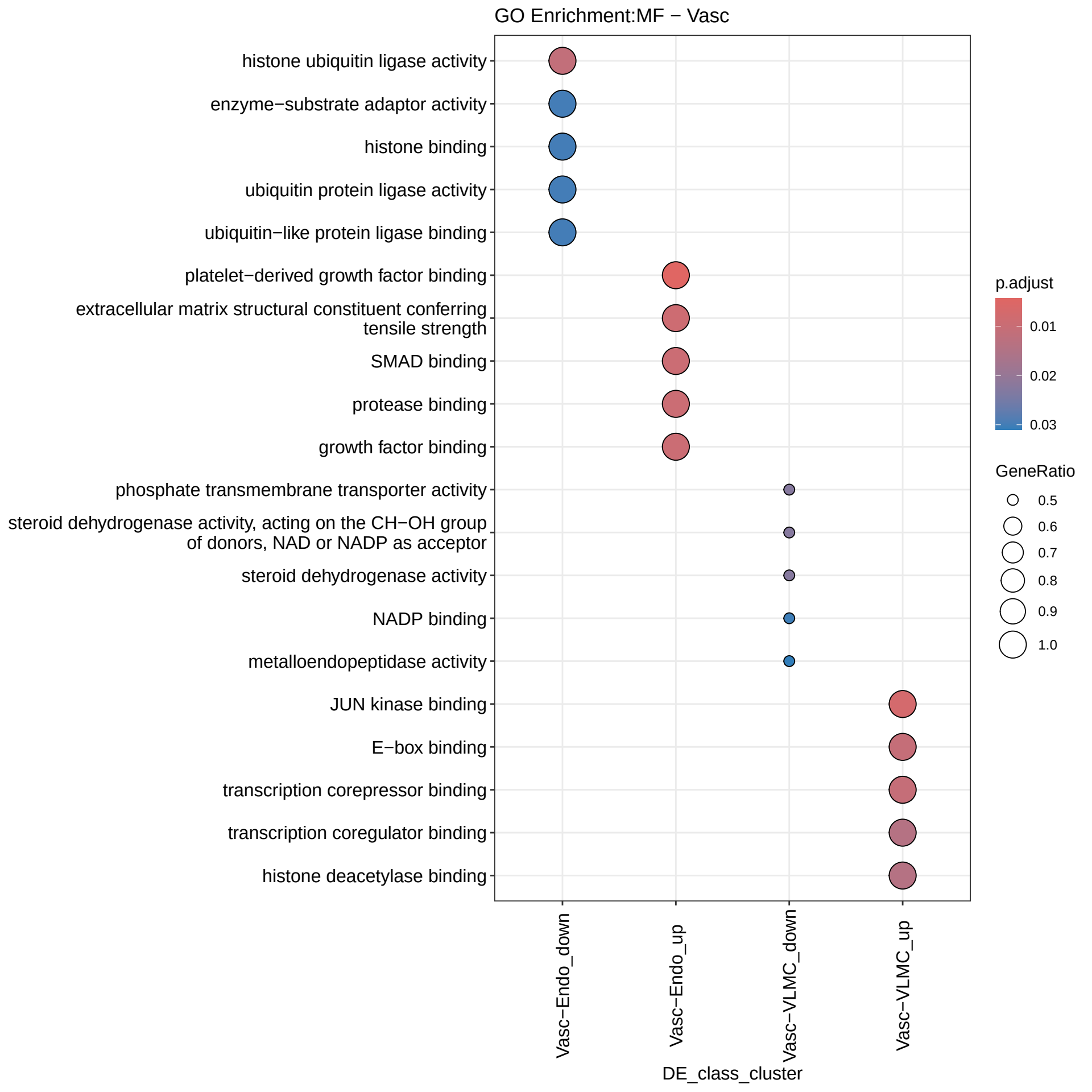

GO Enrichment:CC − Micro
monoatomic ion channel complex
transmembrane transporter complex
transporter complex
nuclear envelope
synaptic membrane
postsynaptic density
asymmetric synapse
postsynaptic specialization
neuron to neuron synapse
Micro−1_up
Micro−3_up
Micro−4_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.4
0.6
0.8
1.0

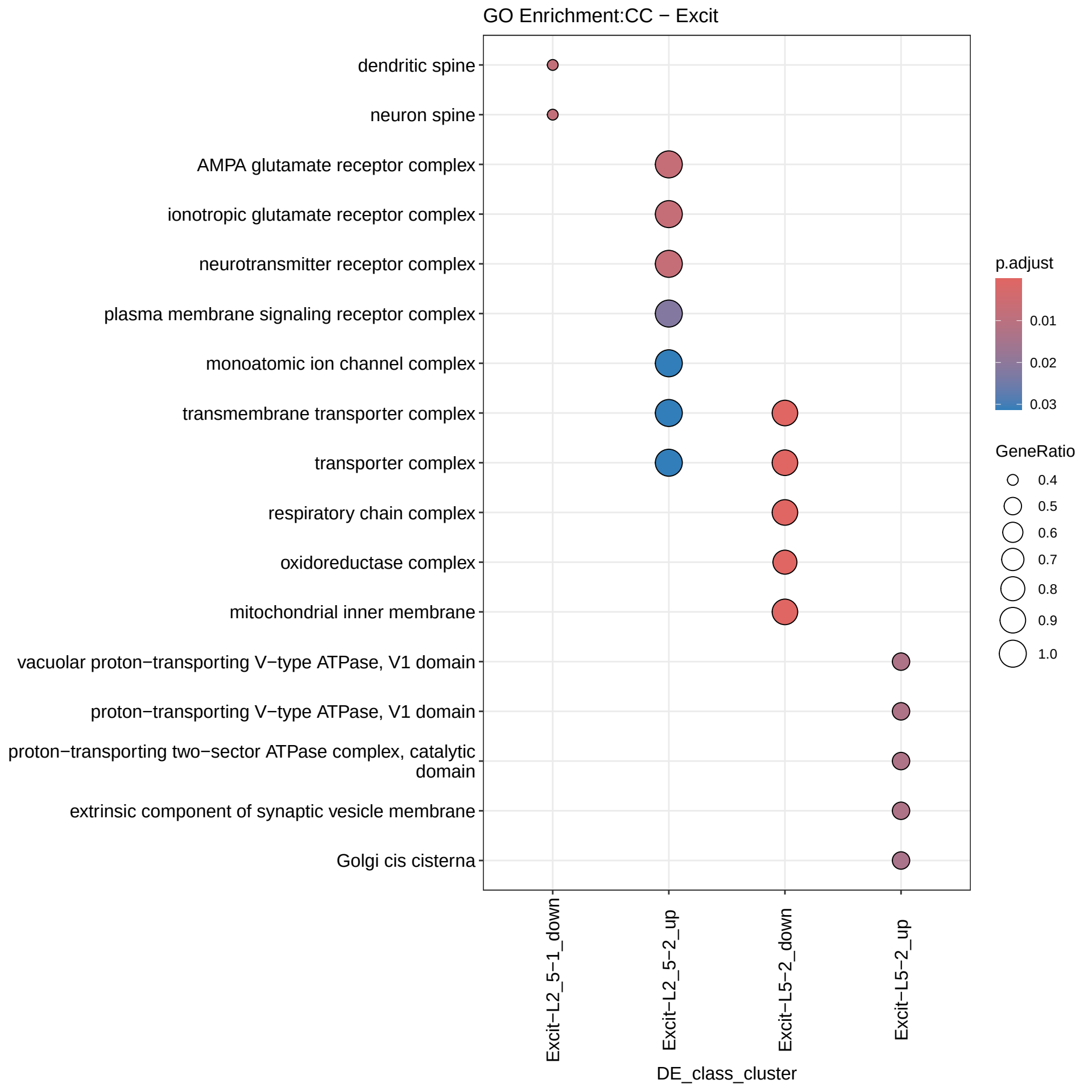

positive regulation of potassium ion transmembrane transport
response to amphetamine
positive regulation of potassium ion transport
response to amine
negative regulation of G protein−coupled receptor signaling pathway
modulation of chemical synaptic transmission
regulation of trans−synaptic signaling

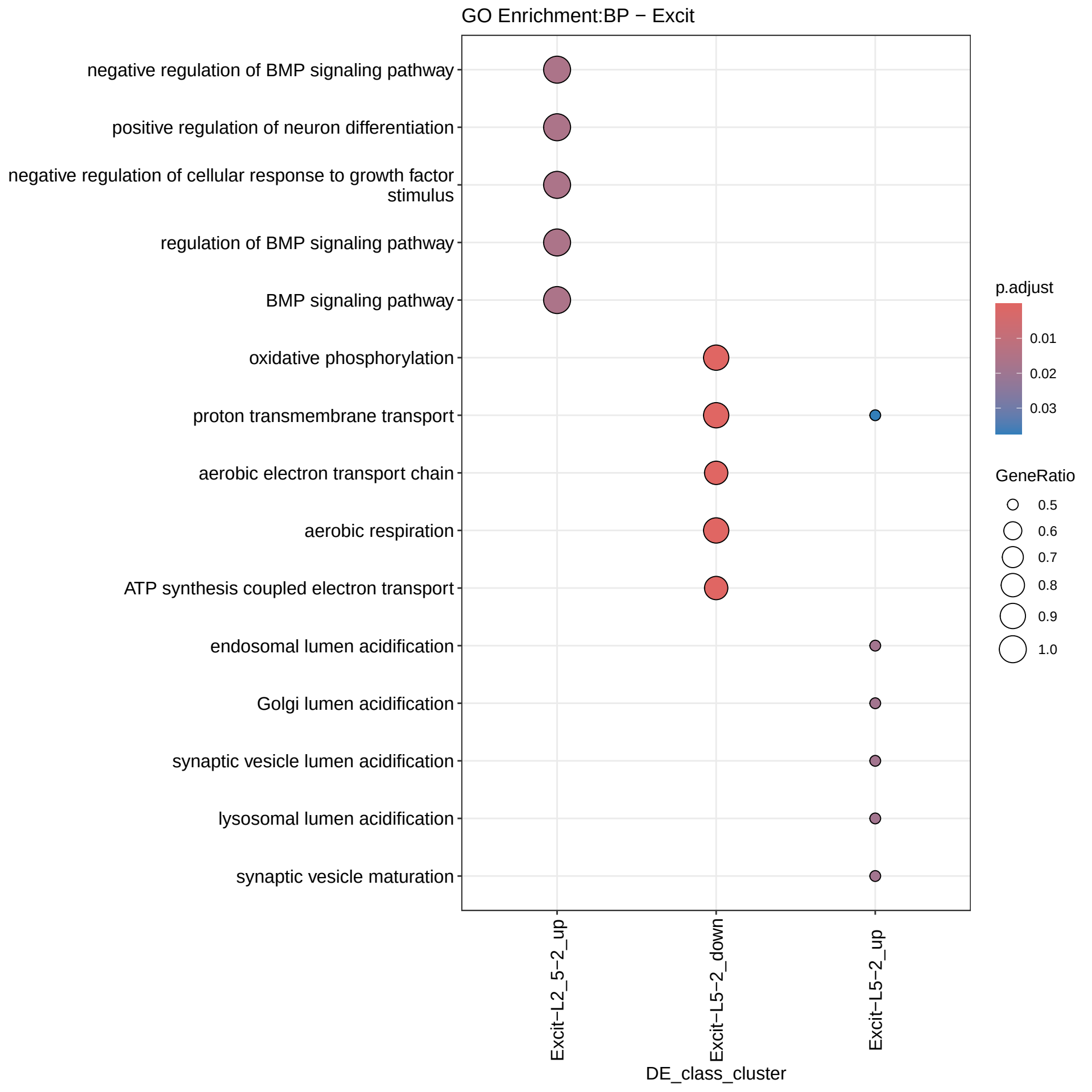

GO Enrichment:MF − Oligo
metal ion transmembrane transporter activity
calmodulin binding
monoatomic cation channel activity
voltage−gated monoatomic cation channel activity
monoatomic ion channel activity
neurexin family protein binding
racemase and epimerase activity
phosphatidic acid binding
ATP hydrolysis activity
phosphatidylinositol−3,4−bisphosphate binding
phosphatidylinositol−3,5−bisphosphate binding
enzyme inhibitor activity
endopeptidase regulator activity
promoter−specific chromatin binding
peptidase regulator activity
Oligo−3_up
Oligo−5_down
Oligo−5_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
0.04
GeneRatio
0.1
0.2
0.3

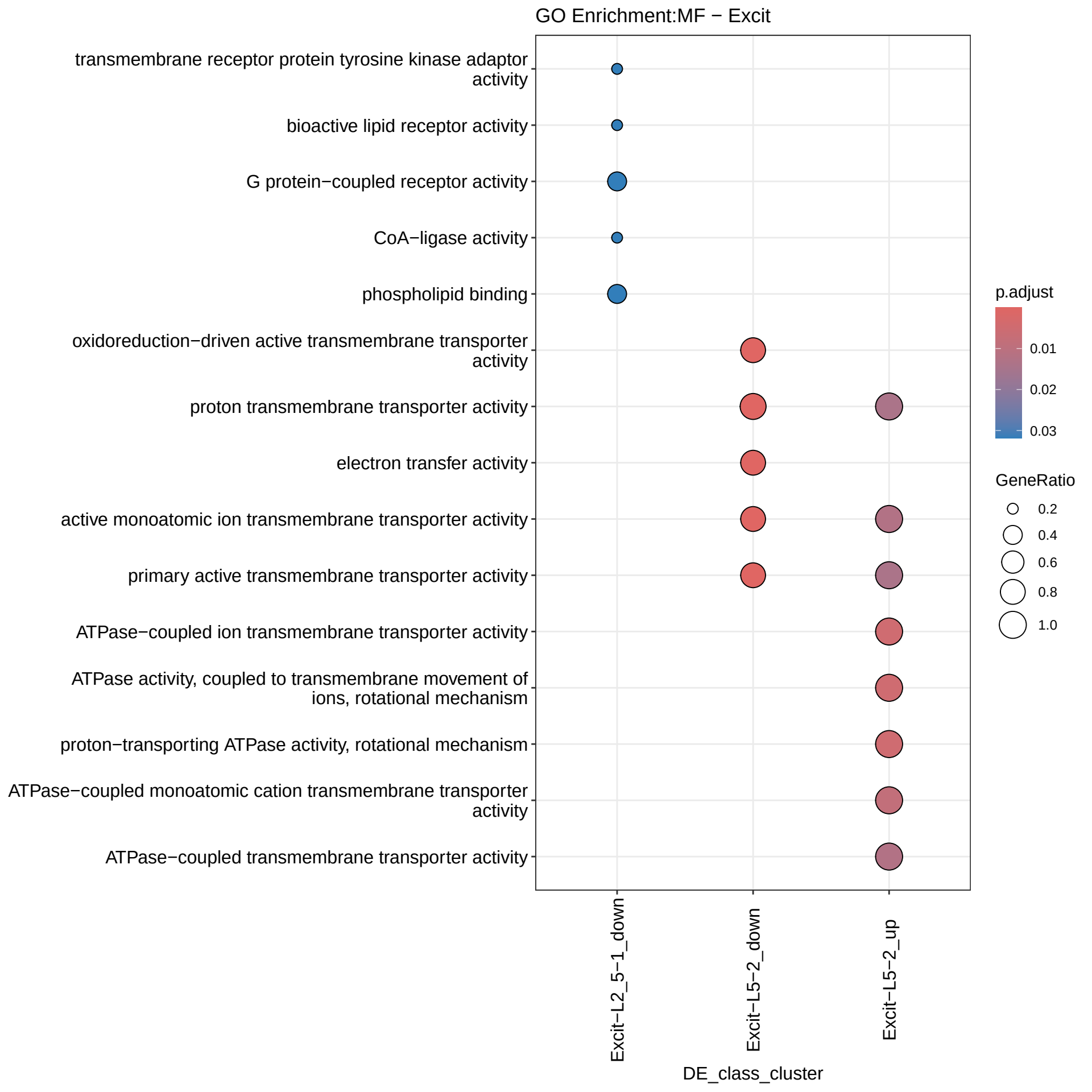

GO Enrichment:CC − Vasc
early endosome membrane
early endosome
lysosomal membrane
lytic vacuole membrane
vacuolar membrane
fibrillar collagen trimer
banded collagen fibril
complex of collagen trimers
collagen trimer
endoplasmic reticulum lumen
nuclear matrix
nuclear periphery
Vasc−Endo_down
Vasc−Endo_up
Vasc−VLMC_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
1

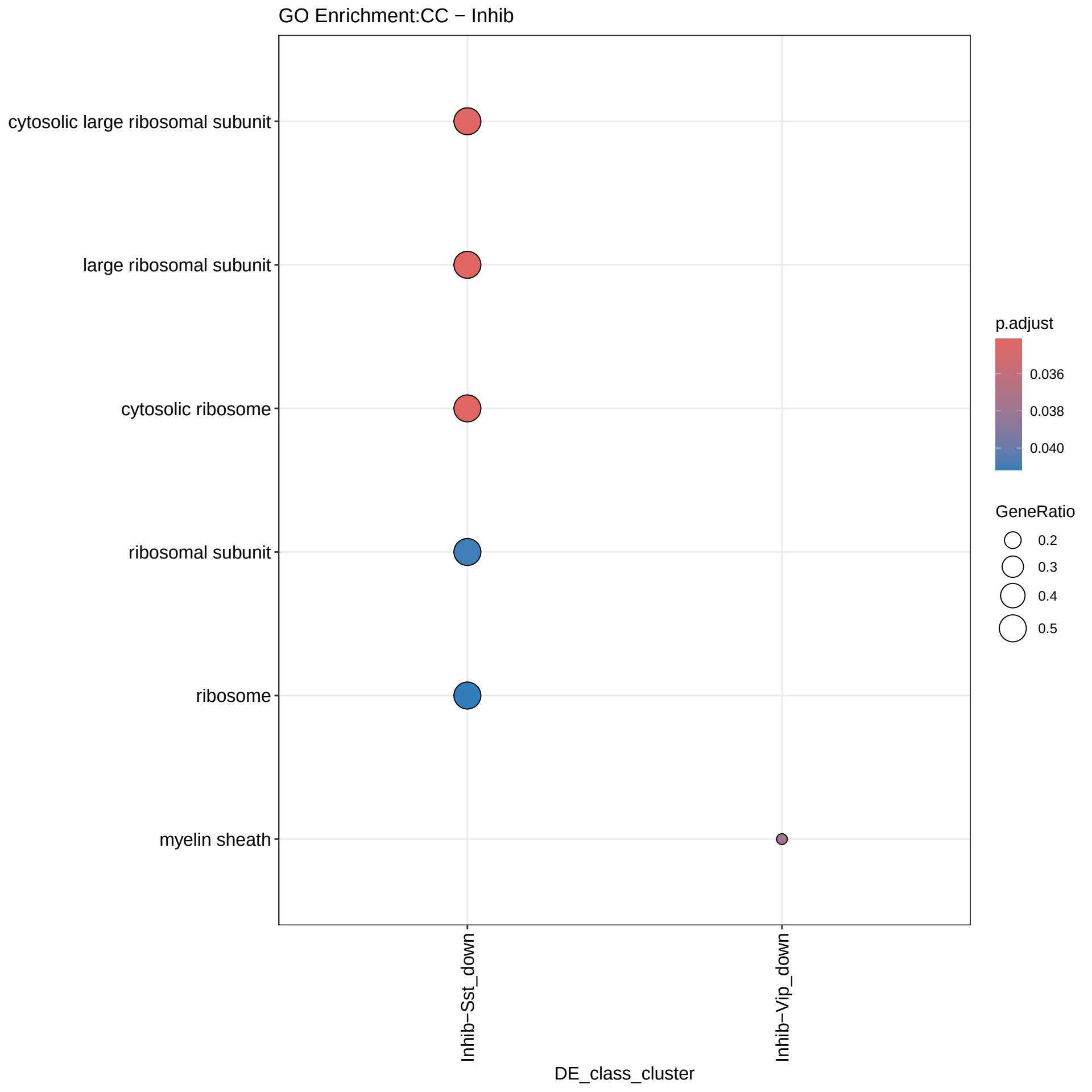

GO Enrichment:BP − Vasc
regulation of protein localization to early endosome
positive regulation of protein localization to early endosome
protein localization to early endosome
regulation of receptor catabolic process
positive regulation of protein localization to endosome
skin morphogenesis
protein trimerization
protein heterooligomerization
extracellular matrix assembly
collagen fibril organization
steroid catabolic process
inorganic anion transport
lung development
respiratory tube development
respiratory system development
negative regulation of epidermal cell differentiation
negative regulation of epidermis development
metanephric tubule morphogenesis
regulation of timing of cell differentiation
positive regulation of astrocyte differentiation
Vasc−Endo_down
Vasc−Endo_up
Vasc−VLMC_down
Vasc−VLMC_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
0.04
GeneRatio
0.5
0.6
0.7
0.8
0.9
1.0

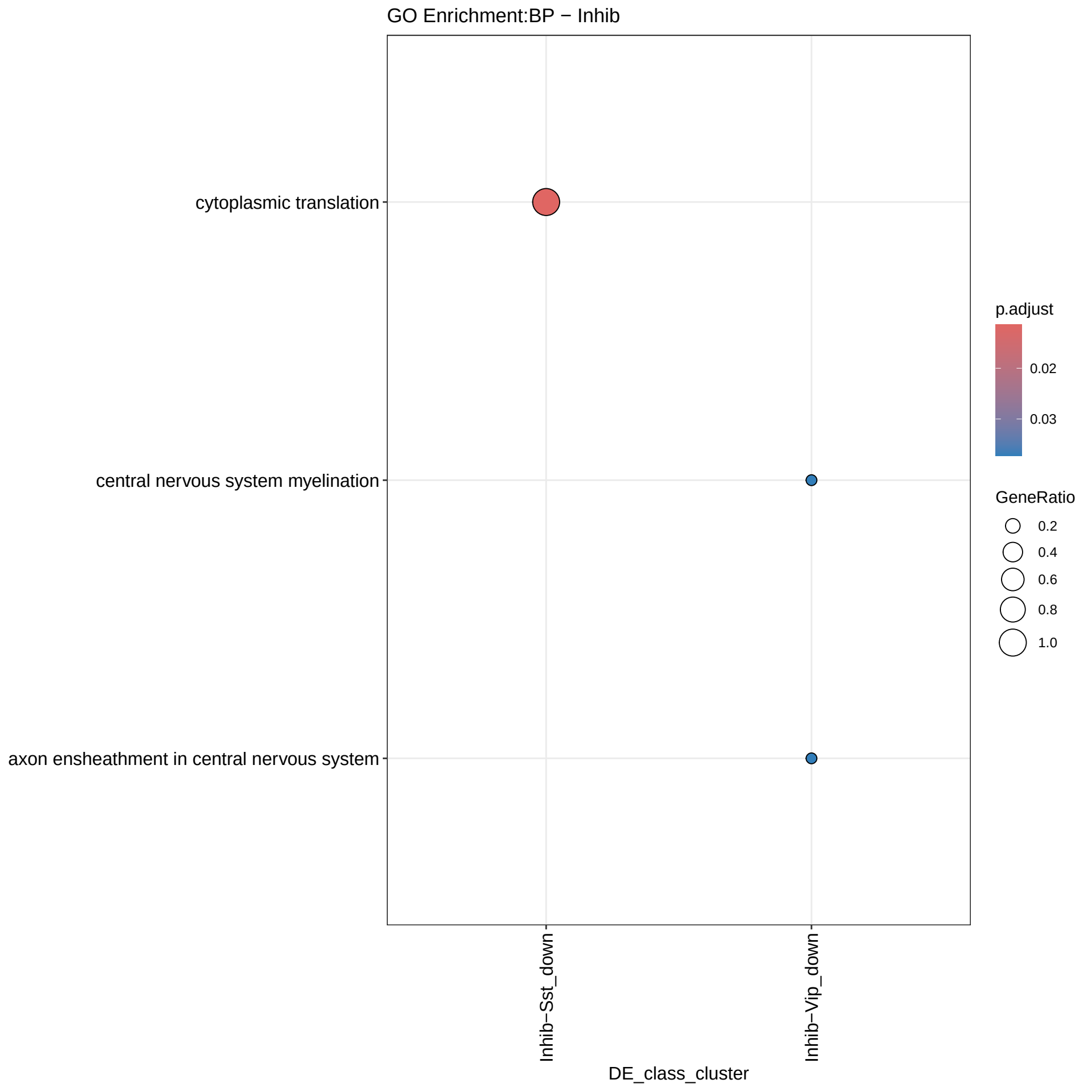

GO Enrichment:MF − Vasc
histone ubiquitin ligase activity
enzyme−substrate adaptor activity
histone binding
ubiquitin protein ligase activity
ubiquitin−like protein ligase binding
platelet−derived growth factor binding
extracellular matrix structural constituent conferring tensile strength
SMAD binding
protease binding
growth factor binding
phosphate transmembrane transporter activity
steroid dehydrogenase activity, acting on the CH−OH group of donors, NAD or NADP as acceptor
steroid dehydrogenase activity
NADP binding
metalloendopeptidase activity
JUN kinase binding
E−box binding
transcription corepressor binding
transcription coregulator binding
histone deacetylase binding
Vasc−Endo_down
Vasc−Endo_up
Vasc−VLMC_down
Vasc−VLMC_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.5
0.6
0.7
0.8
0.9
1.0

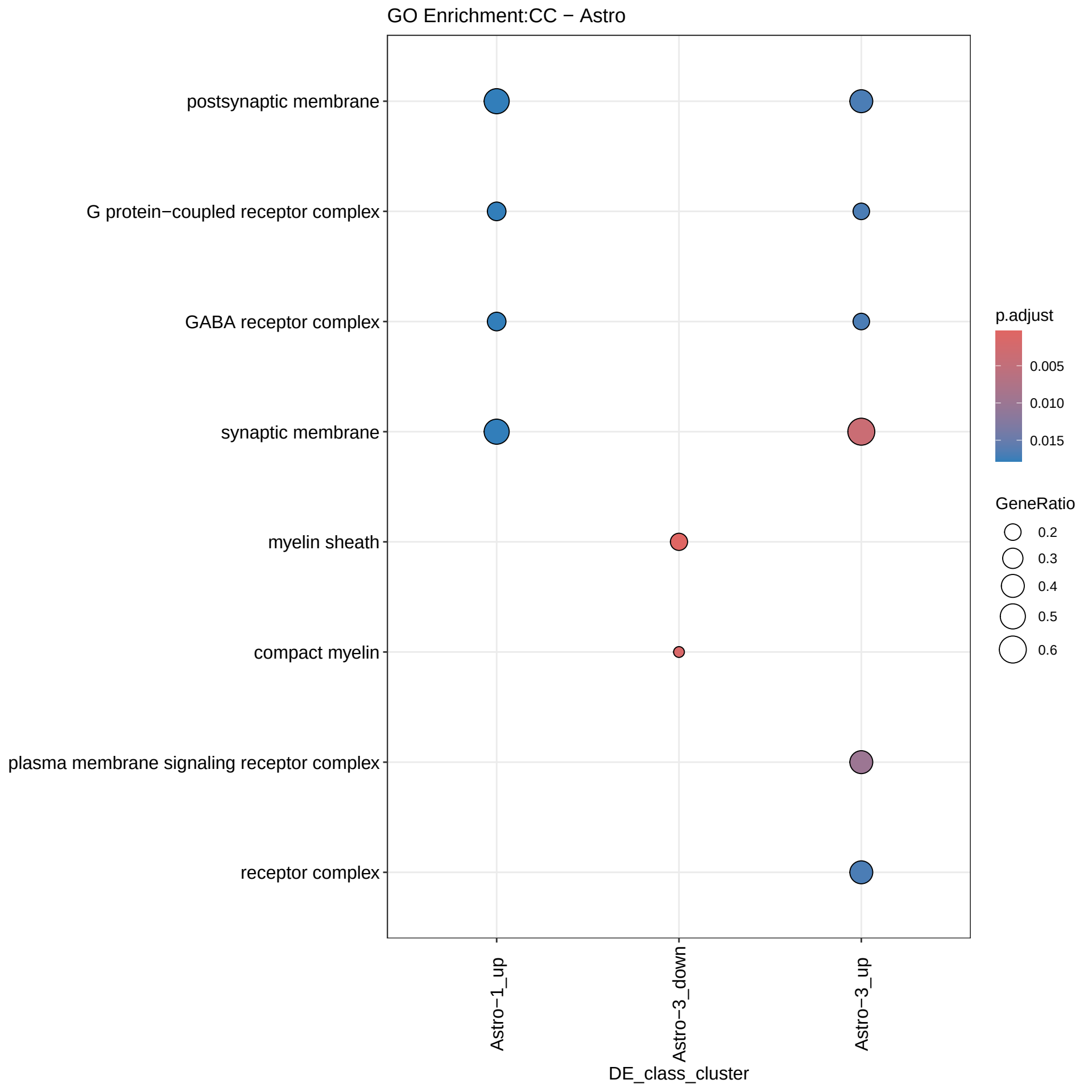

GO Enrichment:CC − Excit
dendritic spine
neuron spine
AMPA glutamate receptor complex
ionotropic glutamate receptor complex
neurotransmitter receptor complex
plasma membrane signaling receptor complex
monoatomic ion channel complex
transmembrane transporter complex
transporter complex
respiratory chain complex
oxidoreductase complex
mitochondrial inner membrane
vacuolar proton−transporting V−type ATPase, V1 domain
proton−transporting V−type ATPase, V1 domain
proton−transporting two−sector ATPase complex, catalytic domain
extrinsic component of synaptic vesicle membrane
Golgi cis cisterna
Excit−L2_5−1_down
Excit−L2_5−2_up
Excit−L5−2_down
Excit−L5−2_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.4
0.5
0.6
0.7
0.8
0.9
1.0

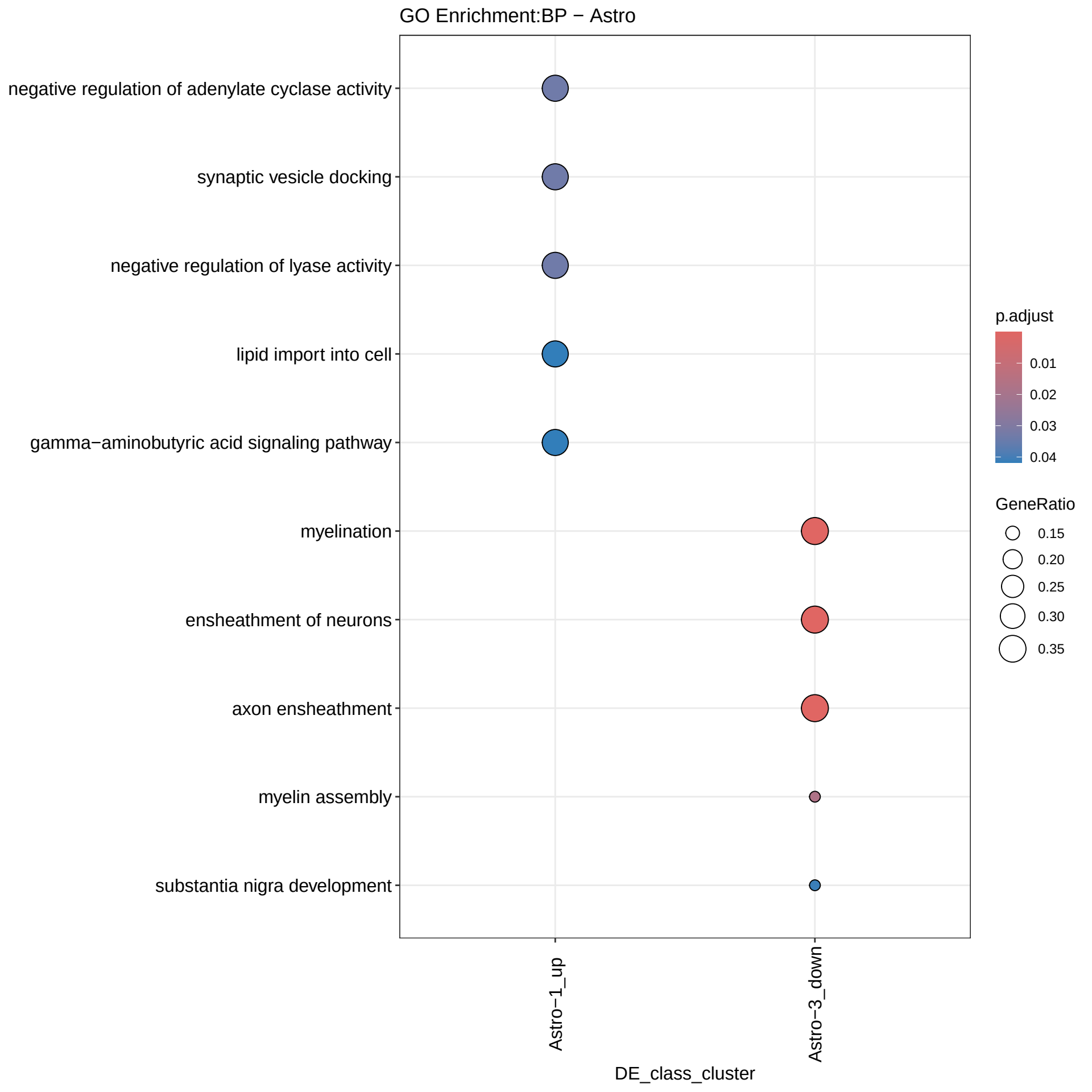

GO Enrichment:BP − Excit
negative regulation of BMP signaling pathway
positive regulation of neuron differentiation
negative regulation of cellular response to growth factor stimulus
regulation of BMP signaling pathway
BMP signaling pathway
oxidative phosphorylation
proton transmembrane transport
aerobic electron transport chain
aerobic respiration
ATP synthesis coupled electron transport
endosomal lumen acidification
Golgi lumen acidification
synaptic vesicle lumen acidification
lysosomal lumen acidification
synaptic vesicle maturation
Excit−L2_5−2_up
Excit−L5−2_down
Excit−L5−2_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.5
0.6
0.7
0.8
0.9
1.0

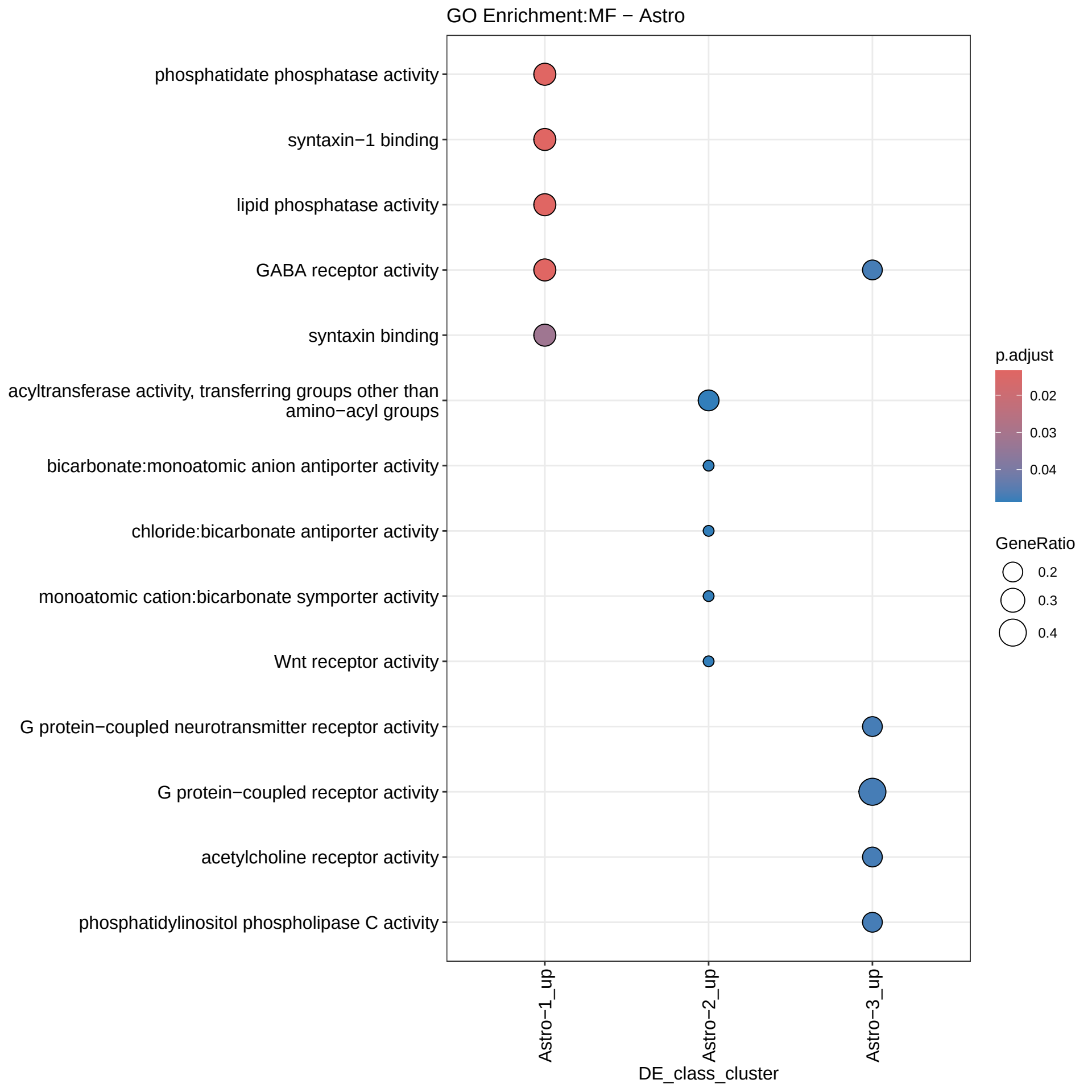

GO Enrichment:MF − Excit
transmembrane receptor protein tyrosine kinase adaptor activity
bioactive lipid receptor activity
G protein−coupled receptor activity
CoA−ligase activity
phospholipid binding
oxidoreduction−driven active transmembrane transporter activity
proton transmembrane transporter activity
electron transfer activity
active monoatomic ion transmembrane transporter activity
primary active transmembrane transporter activity
ATPase−coupled ion transmembrane transporter activity
ATPase activity, coupled to transmembrane movement of ions, rotational mechanism
proton−transporting ATPase activity, rotational mechanism
ATPase−coupled monoatomic cation transmembrane transporter activity
ATPase−coupled transmembrane transporter activity
Excit−L2_5−1_down
Excit−L5−2_down
Excit−L5−2_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.2
0.4
0.6
0.8
1.0

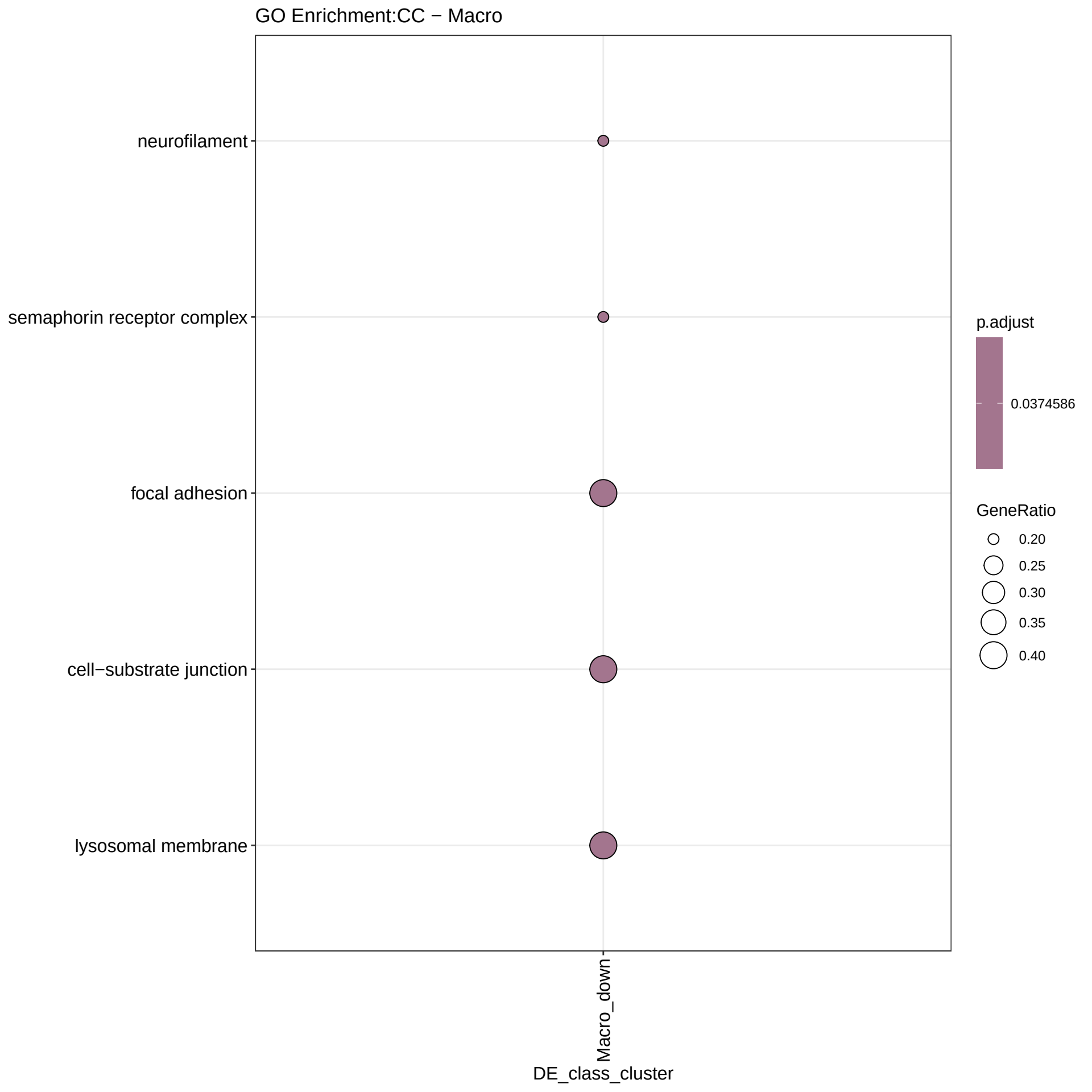

GO Enrichment:CC − Inhib
cytosolic large ribosomal subunit
large ribosomal subunit
cytosolic ribosome
ribosomal subunit
ribosome
myelin sheath
Inhib−Sst_down
Inhib−Vip_down
DE_class_cluster
p.adjust
0.036
0.038
0.040
GeneRatio
0.2
0.3
0.4
0.5

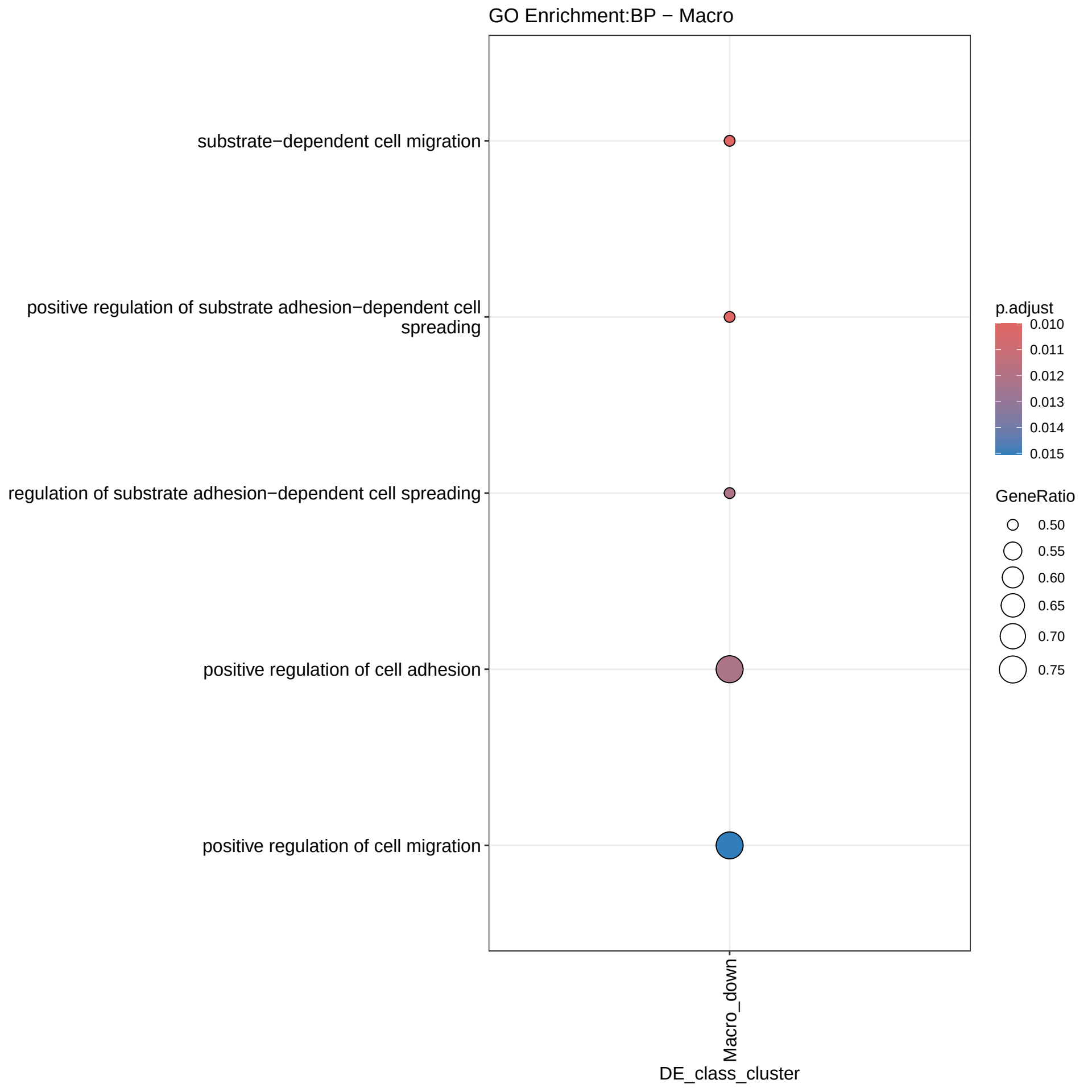

GO Enrichment:BP − Inhib
cytoplasmic translation
central nervous system myelination
axon ensheathment in central nervous system
p.adjust
0.02
0.03
GeneRatio
0.2
0.4
0.6
0.8
1.0
Inhib−Sst_down
Inhib−Vip_down
DE_class_cluster

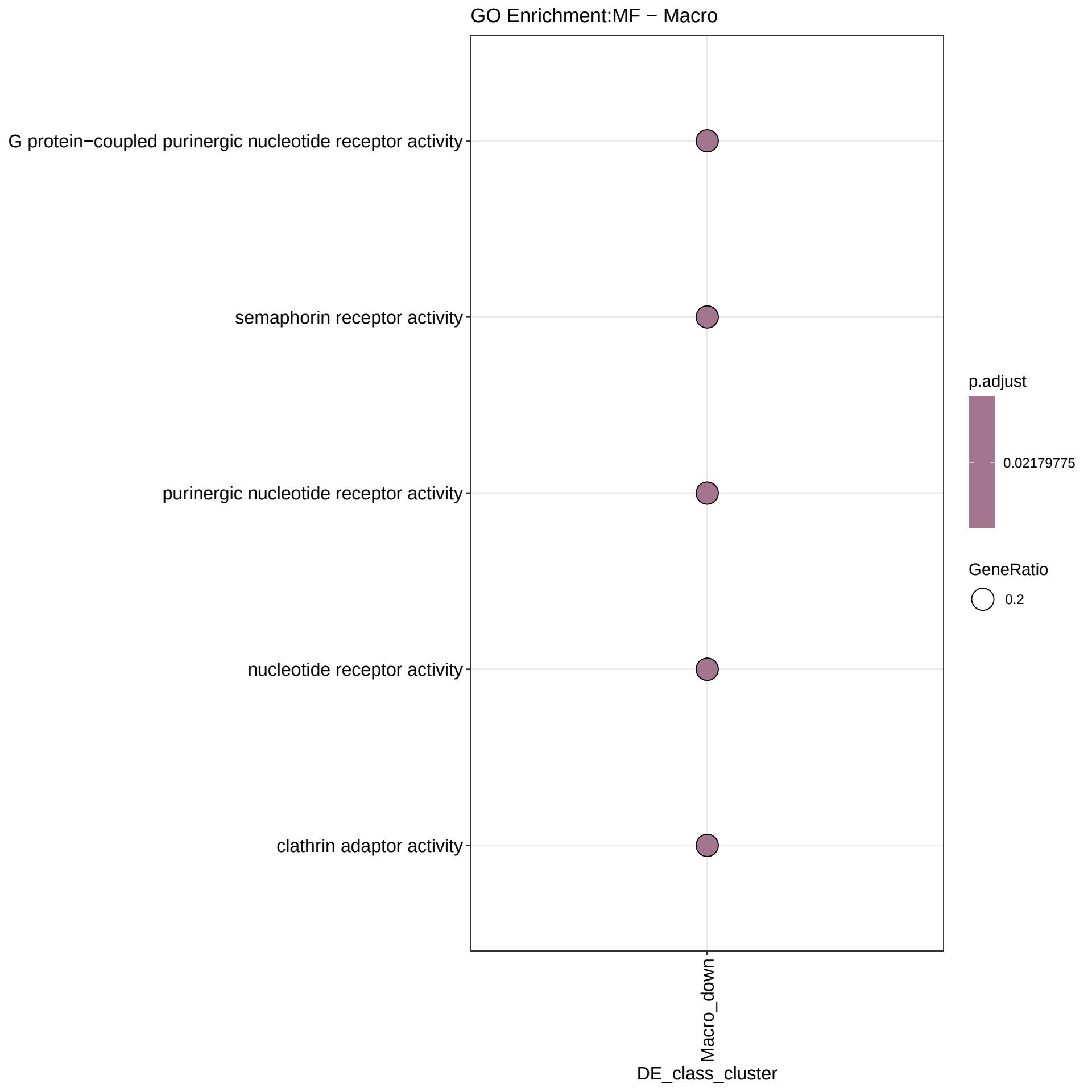

GO Enrichment:CC − Astro
postsynaptic membrane
G protein−coupled receptor complex
GABA receptor complex
synaptic membrane
myelin sheath
compact myelin
plasma membrane signaling receptor complex
receptor complex
Astro−1_up
Astro−3_down
Astro−3_up
DE_class_cluster
p.adjust
0.005
0.010
0.015
GeneRatio
0.2
0.3
0.4
0.5
0.6

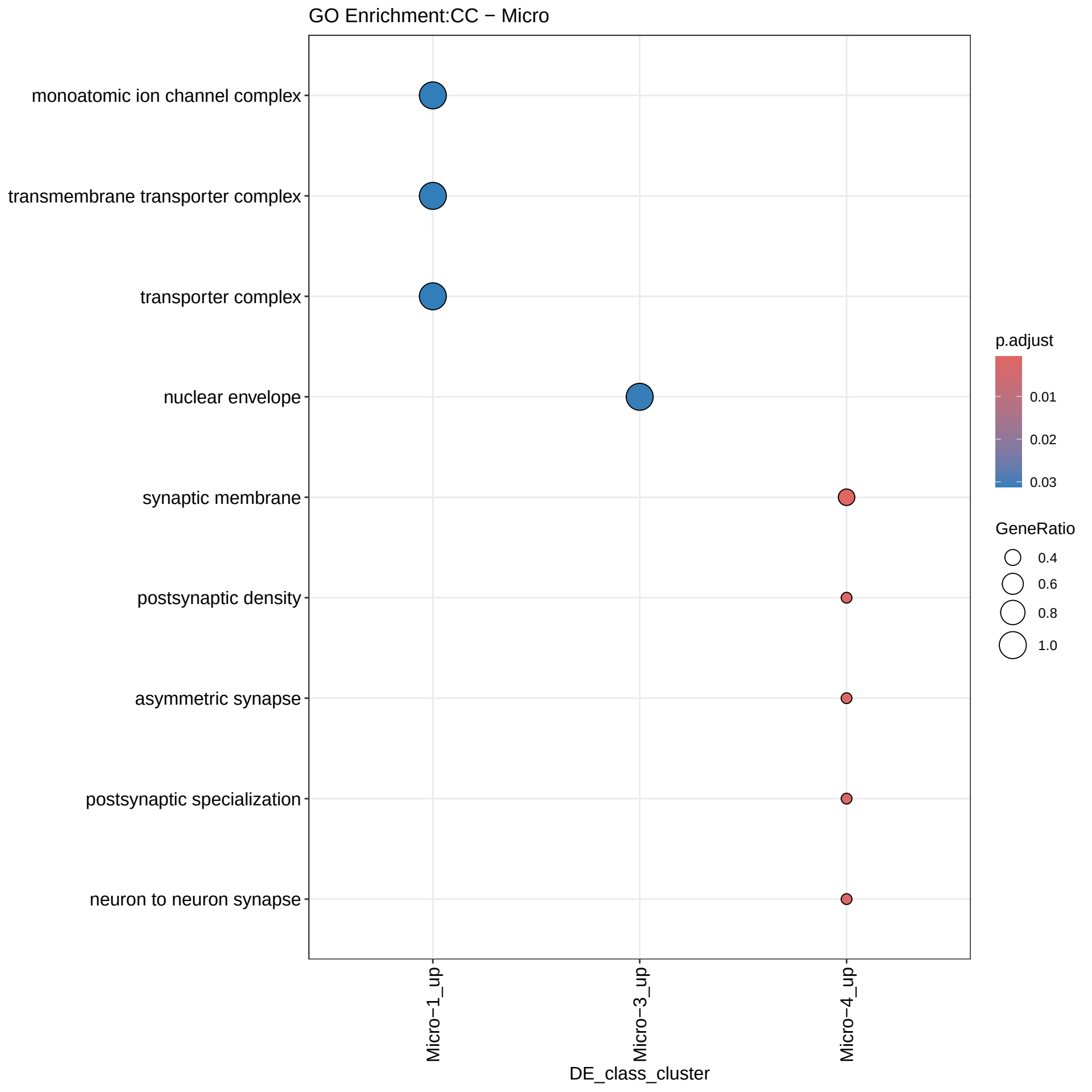

GO Enrichment:BP − Astro
negative regulation of adenylate cyclase activity
synaptic vesicle docking
negative regulation of lyase activity
lipid import into cell
gamma−aminobutyric acid signaling pathway
myelination
ensheathment of neurons
axon ensheathment
myelin assembly
substantia nigra development
Astro−1_up
Astro−3_down
DE_class_cluster
p.adjust
0.01
0.02
0.03
0.04
GeneRatio
0.15
0.20
0.25
0.30
0.35

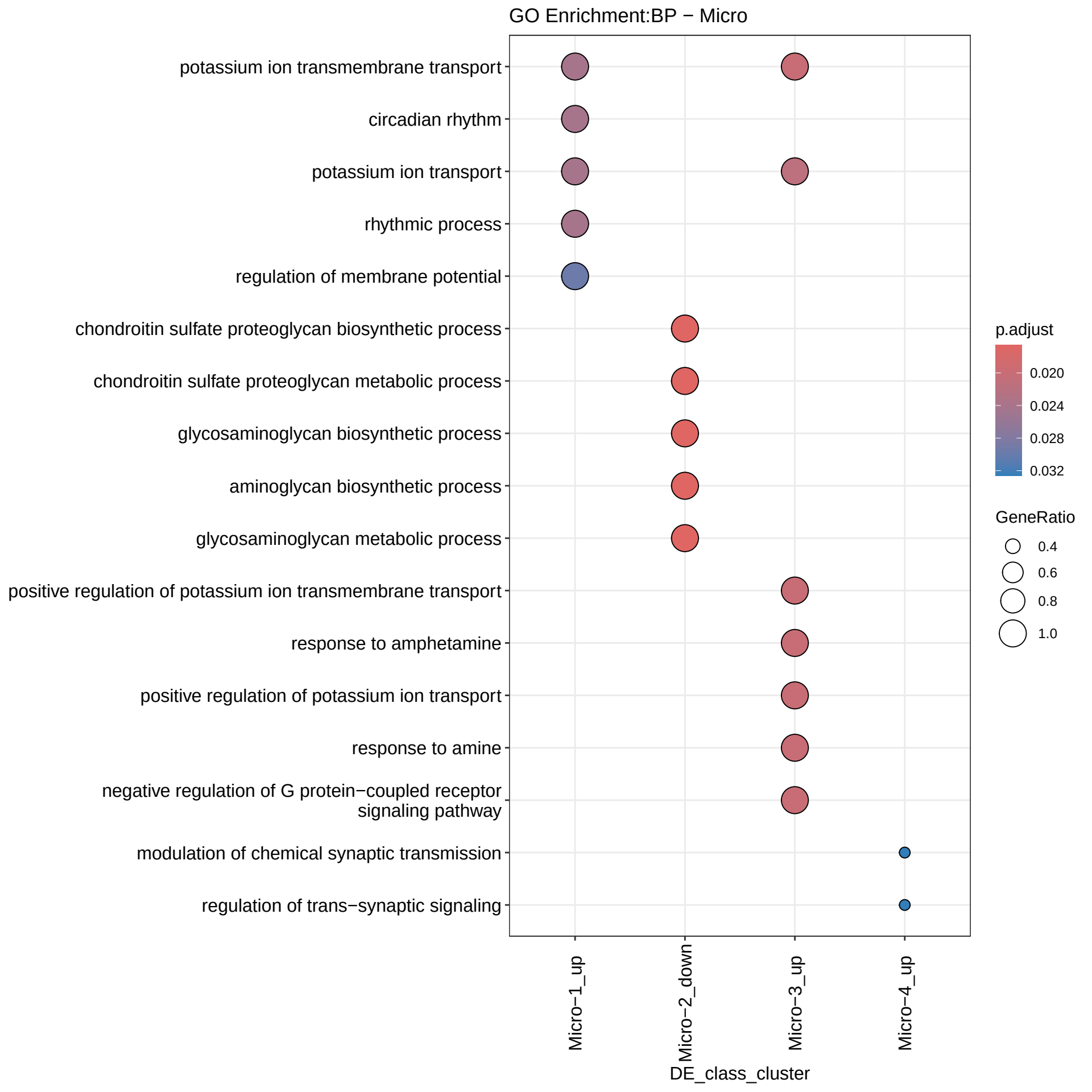

GO Enrichment:MF − Astro
phosphatidate phosphatase activity
syntaxin−1 binding
lipid phosphatase activity
GABA receptor activity
syntaxin binding
acyltransferase activity, transferring groups other than amino−acyl groups
bicarbonate:monoatomic anion antiporter activity
chloride:bicarbonate antiporter activity
monoatomic cation:bicarbonate symporter activity
Wnt receptor activity
G protein−coupled neurotransmitter receptor activity
G protein−coupled receptor activity
acetylcholine receptor activity
phosphatidylinositol phospholipase C activity
Astro−1_up
Astro−2_up
Astro−3_up
DE_class_cluster
p.adjust
0.02
0.03
0.04
GeneRatio
0.2
0.3
0.4

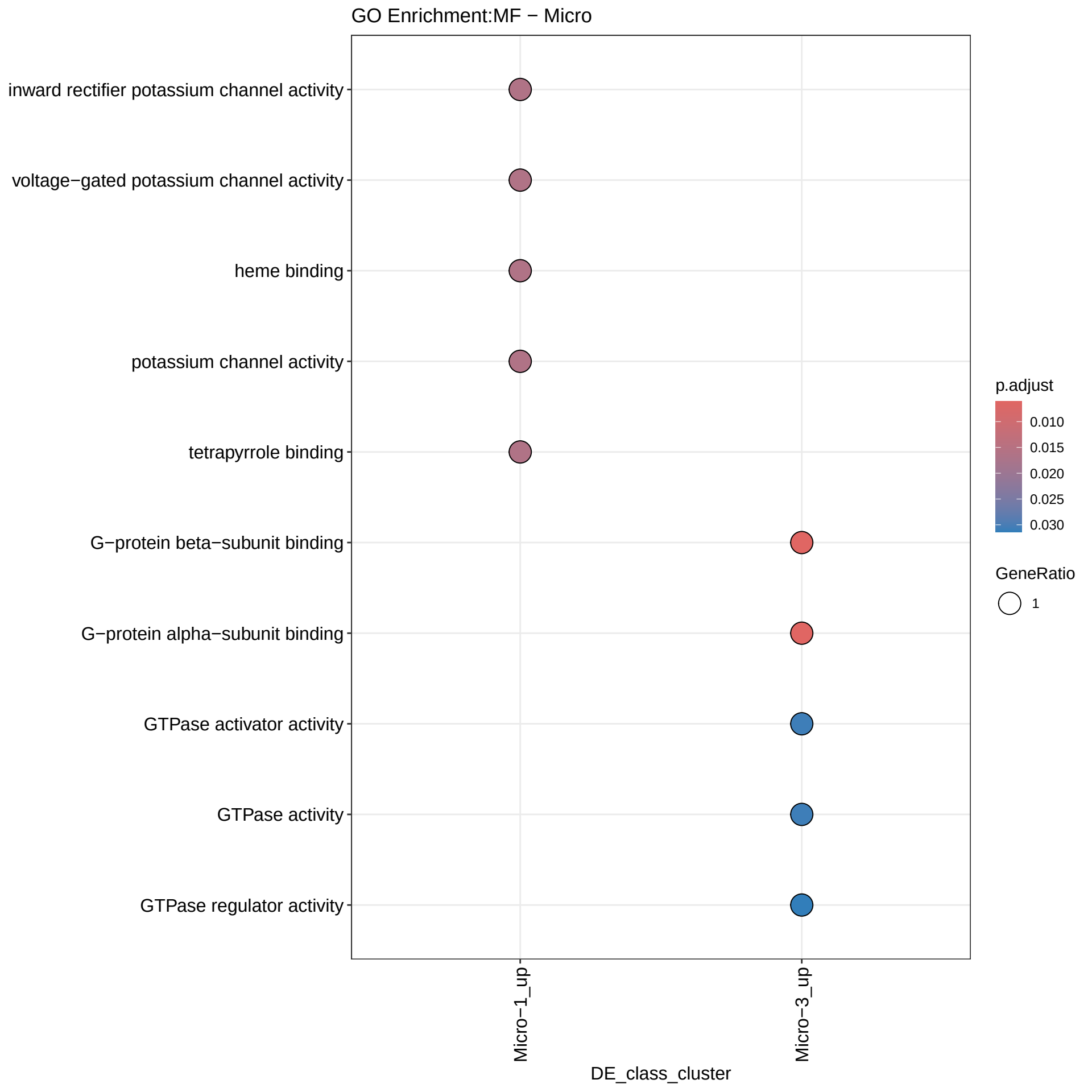

GO Enrichment:CC − Macro
neurofilament
semaphorin receptor complex
focal adhesion
cell−substrate junction
lysosomal membrane
Macro_down
DE_class_cluster
p.adjust
0.0374586
GeneRatio
0.20
0.25
0.30
0.35
0.40

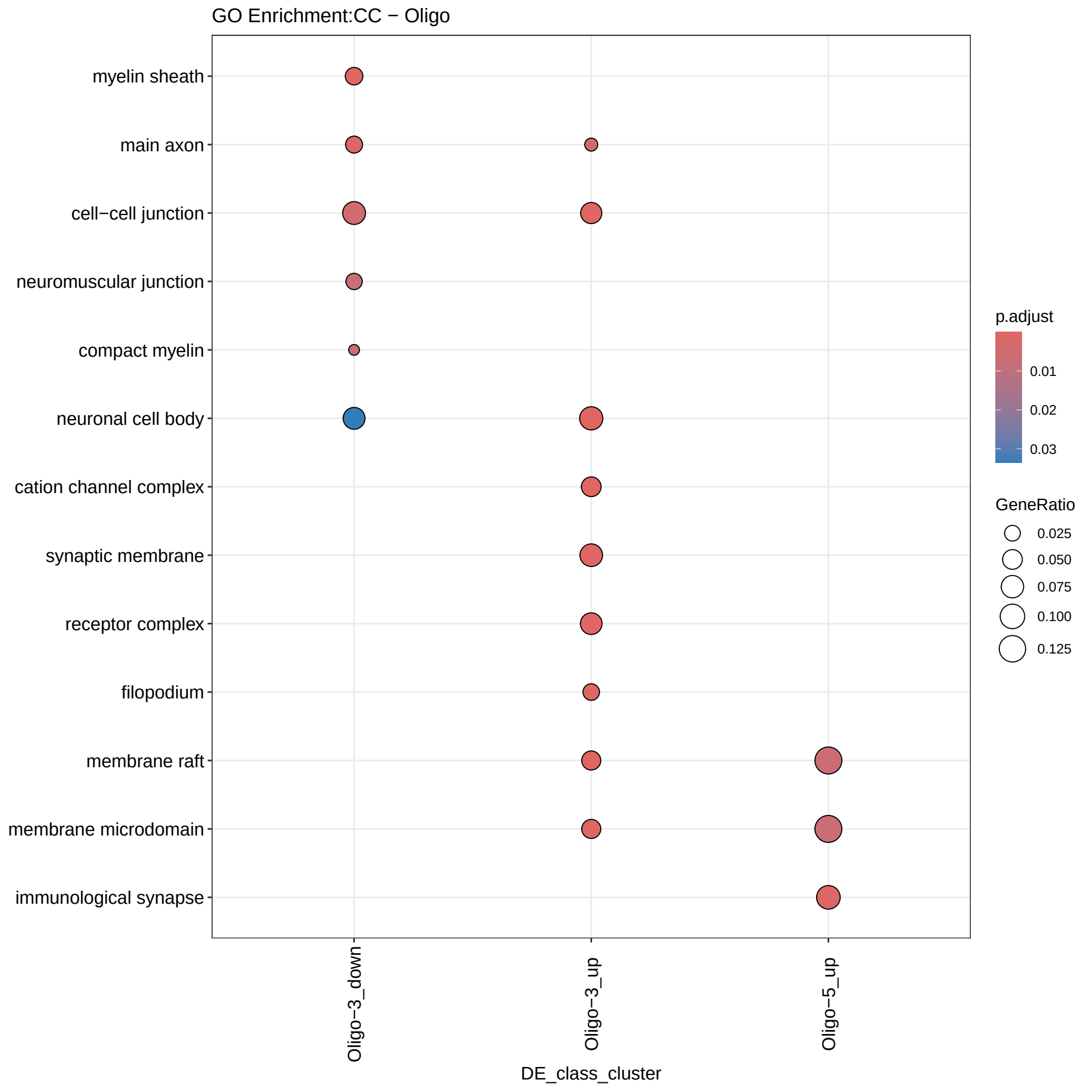

GO Enrichment:BP − Macro
substrate−dependent cell migration
positive regulation of substrate adhesion−dependent cell spreading
regulation of substrate adhesion−dependent cell spreading
positive regulation of cell adhesion
positive regulation of cell migration
Macro_down
DE_class_cluster
p.adjust
0.010
0.011
0.012
0.013
0.014
0.015
GeneRatio
0.50
0.55
0.60
0.65
0.70
0.75

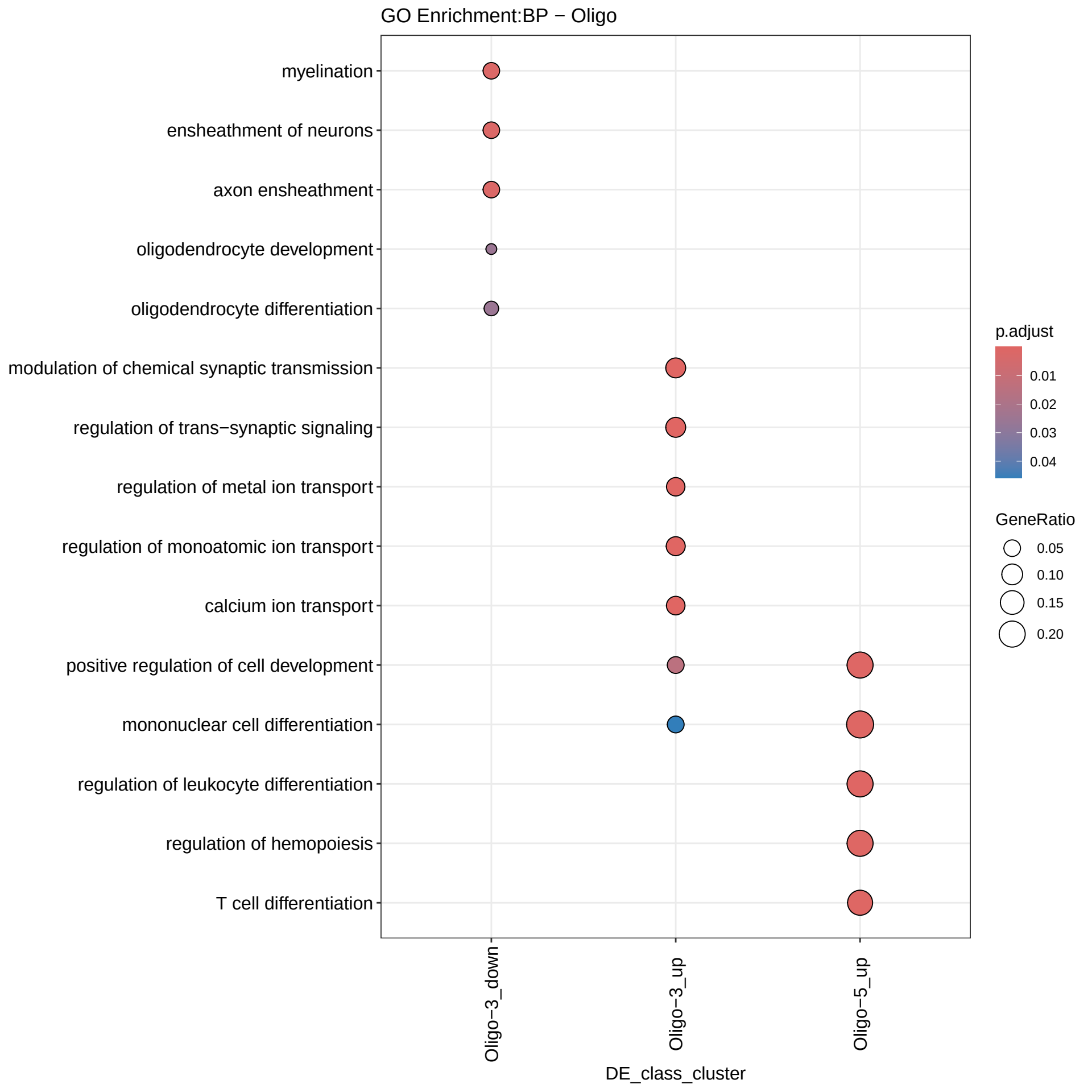

GO Enrichment:MF − Macro
G protein−coupled purinergic nucleotide receptor activity
semaphorin receptor activity
purinergic nucleotide receptor activity
nucleotide receptor activity
clathrin adaptor activity
Macro_down
DE_class_cluster
p.adjust
0.02179775
GeneRatio
0.2

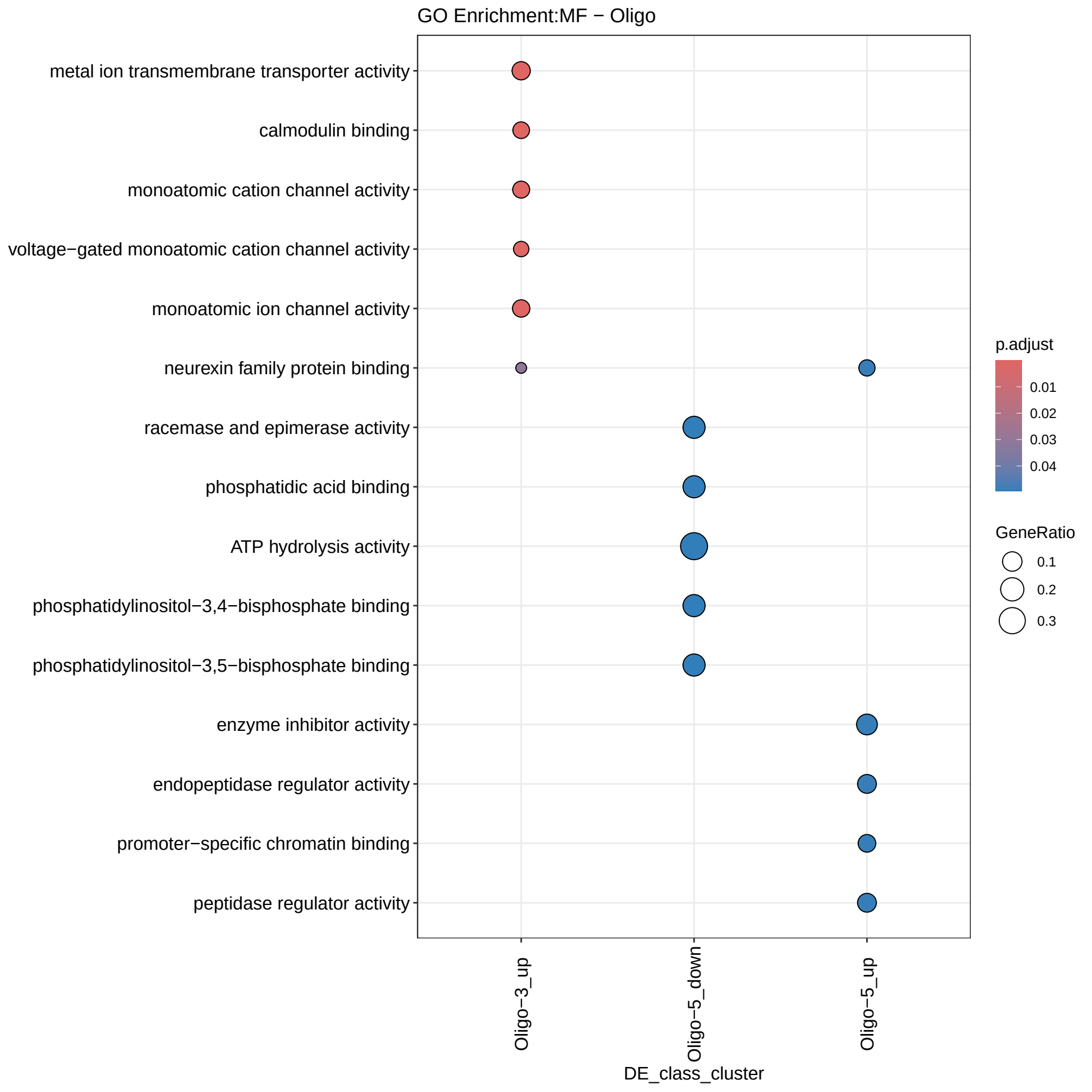

GO Enrichment:CC − Micro
monoatomic ion channel complex
transmembrane transporter complex
transporter complex
nuclear envelope
synaptic membrane
postsynaptic density
asymmetric synapse
postsynaptic specialization
neuron to neuron synapse
Micro−1_up
Micro−3_up
Micro−4_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.4
0.6
0.8
1.0

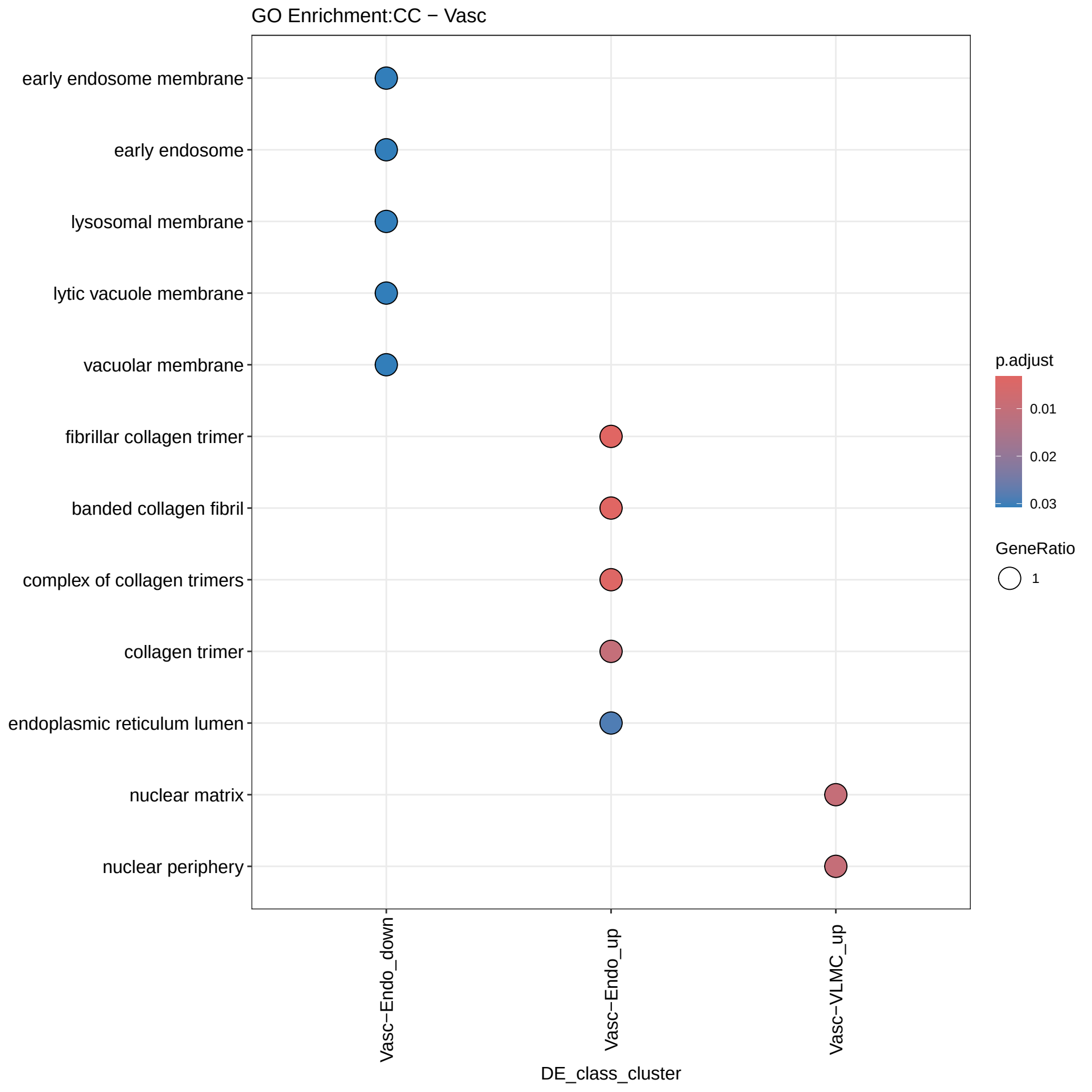

GO Enrichment:BP − Micro
potassium ion transmembrane transport
circadian rhythm
potassium ion transport
rhythmic process
regulation of membrane potential
chondroitin sulfate proteoglycan biosynthetic process
chondroitin sulfate proteoglycan metabolic process
glycosaminoglycan biosynthetic process
aminoglycan biosynthetic process
glycosaminoglycan metabolic process
positive regulation of potassium ion transmembrane transport
response to amphetamine
positive regulation of potassium ion transport
response to amine
negative regulation of G protein−coupled receptor signaling pathway
modulation of chemical synaptic transmission
regulation of trans−synaptic signaling
Micro−1_up
Micro−2_down
Micro−3_up
Micro−4_up
DE_class_cluster
p.adjust
0.020
0.024
0.028
0.032
GeneRatio
0.4
0.6
0.8
1.0

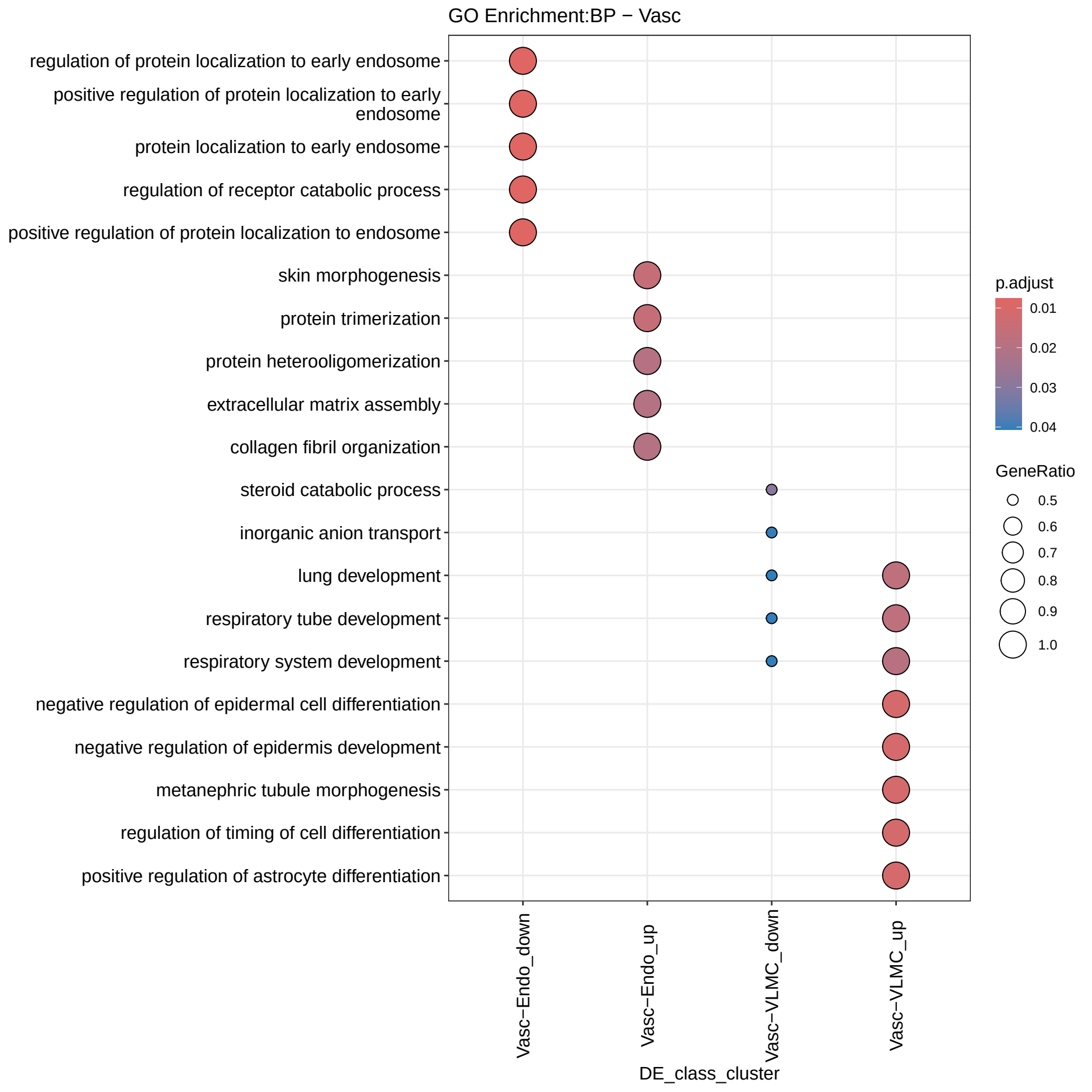

GO Enrichment:MF − Micro
inward rectifier potassium channel activity
voltage−gated potassium channel activity
heme binding
potassium channel activity
tetrapyrrole binding
G−protein beta−subunit binding
G−protein alpha−subunit binding
GTPase activator activity
GTPase activity
GTPase regulator activity
Micro−1_up
Micro−3_up
DE_class_cluster
p.adjust
0.010
0.015
0.020
0.025
0.030
GeneRatio
1

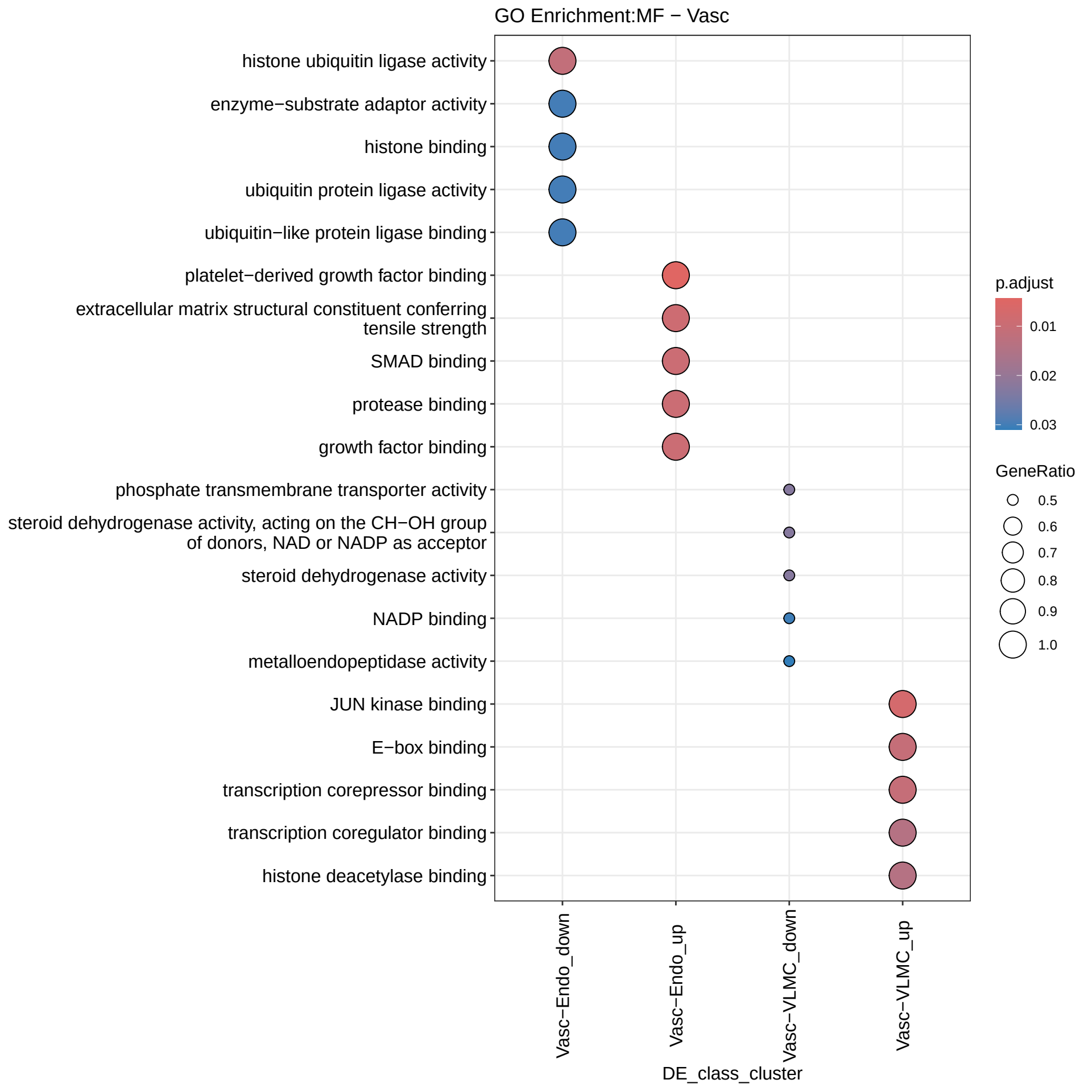

GO Enrichment:CC − Oligo
myelin sheath
main axon
cell−cell junction
neuromuscular junction
compact myelin
neuronal cell body
cation channel complex
synaptic membrane
receptor complex
filopodium
membrane raft
membrane microdomain
immunological synapse
Oligo−3_down
Oligo−3_up
Oligo−5_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.025
0.050
0.075
0.100
0.125

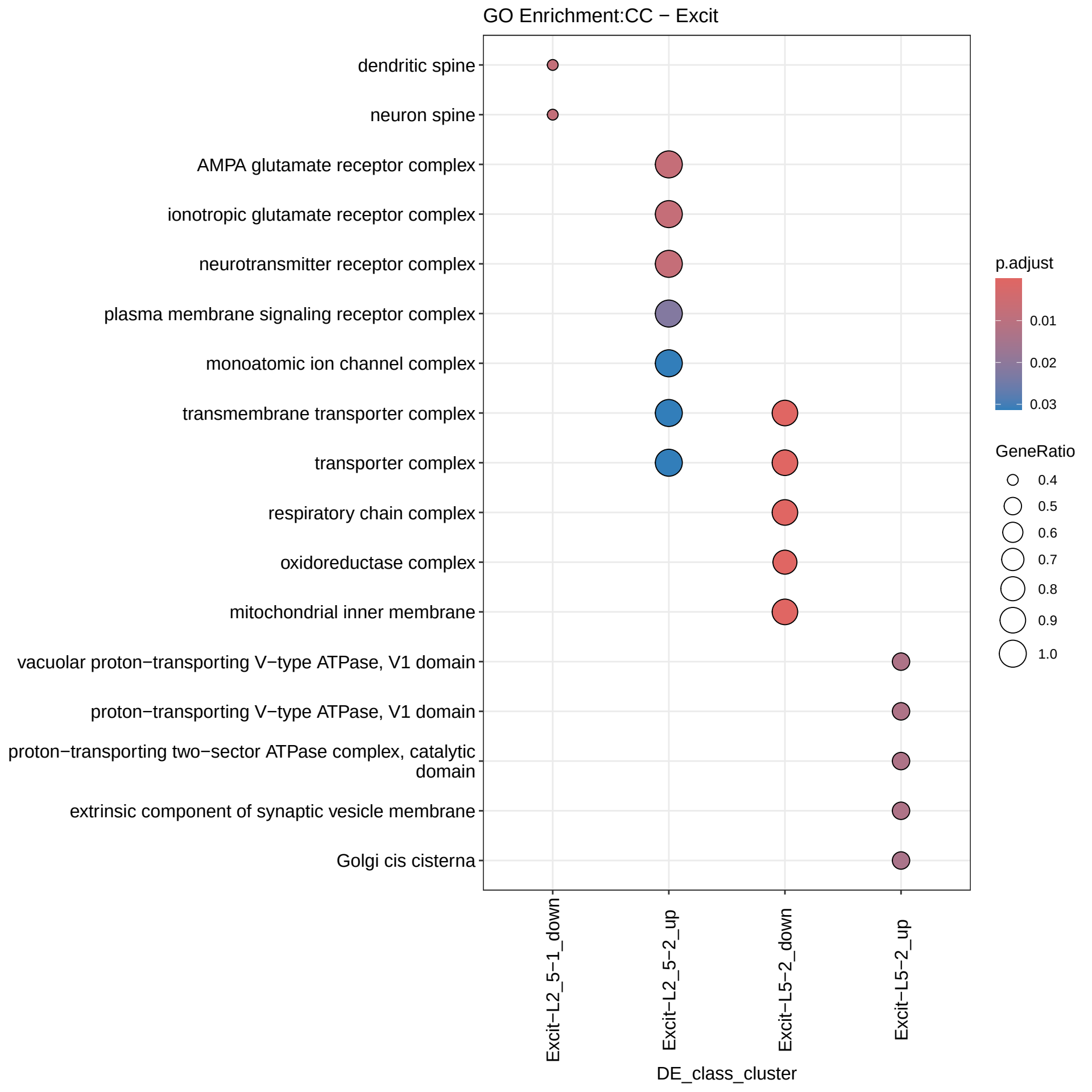

GO Enrichment:BP − Oligo
myelination
ensheathment of neurons
axon ensheathment
oligodendrocyte development
oligodendrocyte differentiation
modulation of chemical synaptic transmission
regulation of trans−synaptic signaling
regulation of metal ion transport
regulation of monoatomic ion transport
calcium ion transport
positive regulation of cell development
mononuclear cell differentiation
regulation of leukocyte differentiation
regulation of hemopoiesis
T cell differentiation
Oligo−3_down
Oligo−3_up
Oligo−5_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
0.04
GeneRatio
0.05
0.10
0.15
0.20

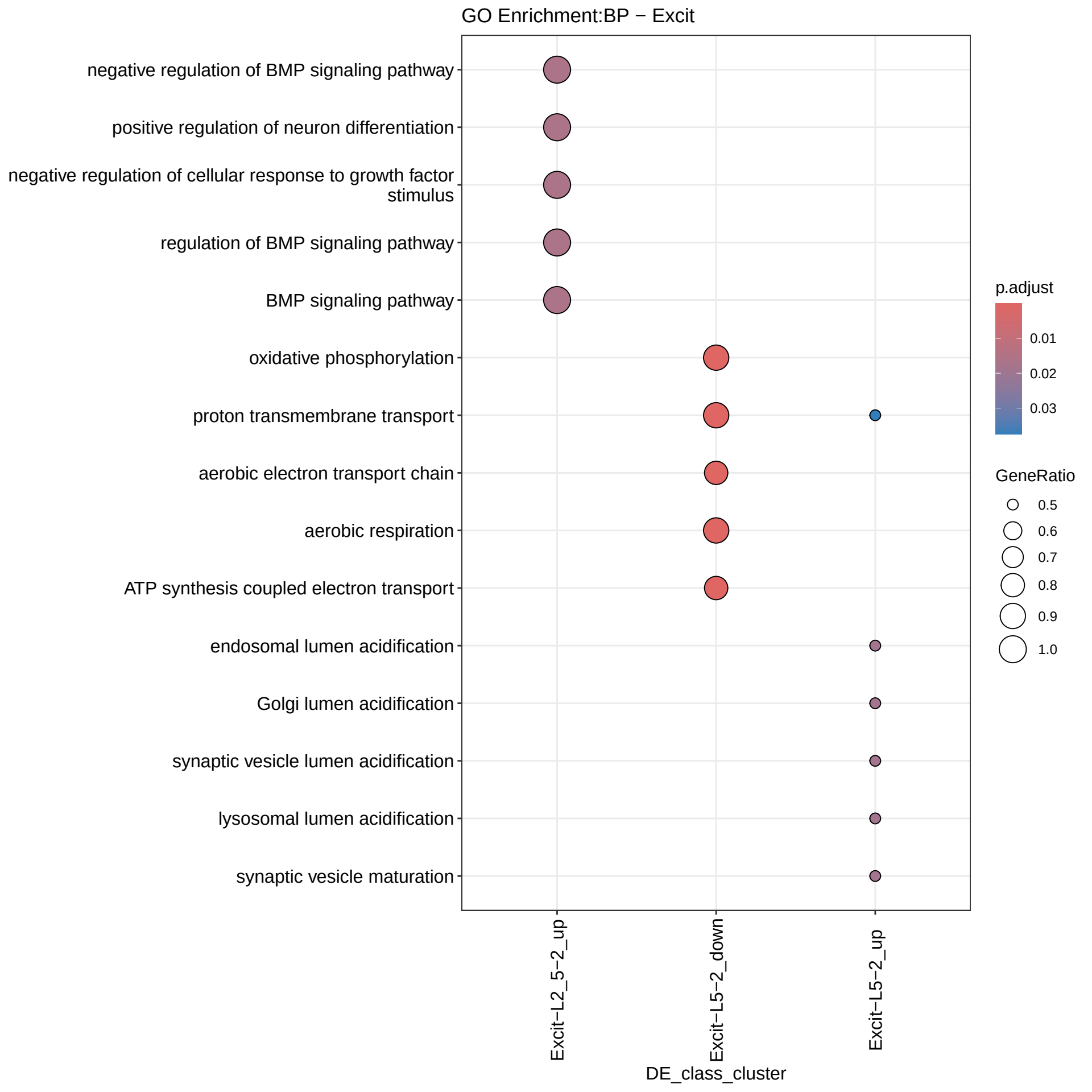

GO Enrichment:MF − Oligo
metal ion transmembrane transporter activity
calmodulin binding
monoatomic cation channel activity
voltage−gated monoatomic cation channel activity
monoatomic ion channel activity
neurexin family protein binding
racemase and epimerase activity
phosphatidic acid binding
ATP hydrolysis activity
phosphatidylinositol−3,4−bisphosphate binding
phosphatidylinositol−3,5−bisphosphate binding
enzyme inhibitor activity
endopeptidase regulator activity
promoter−specific chromatin binding
peptidase regulator activity
Oligo−3_up
Oligo−5_down
Oligo−5_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
0.04
GeneRatio
0.1
0.2
0.3

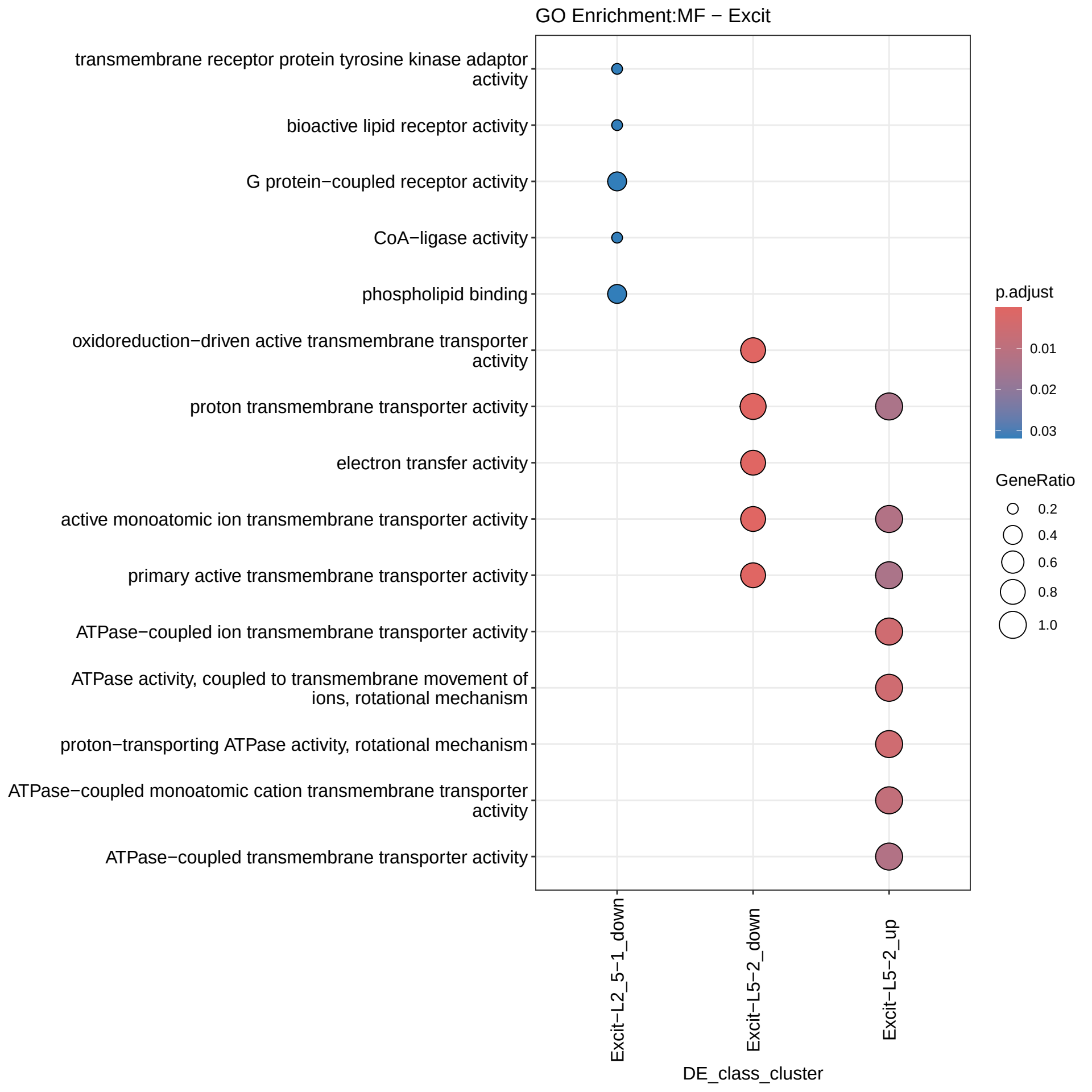

GO Enrichment:CC − Vasc
early endosome membrane
early endosome
lysosomal membrane
lytic vacuole membrane
vacuolar membrane
fibrillar collagen trimer
banded collagen fibril
complex of collagen trimers
collagen trimer
endoplasmic reticulum lumen
nuclear matrix
nuclear periphery
Vasc−Endo_down
Vasc−Endo_up
Vasc−VLMC_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
1

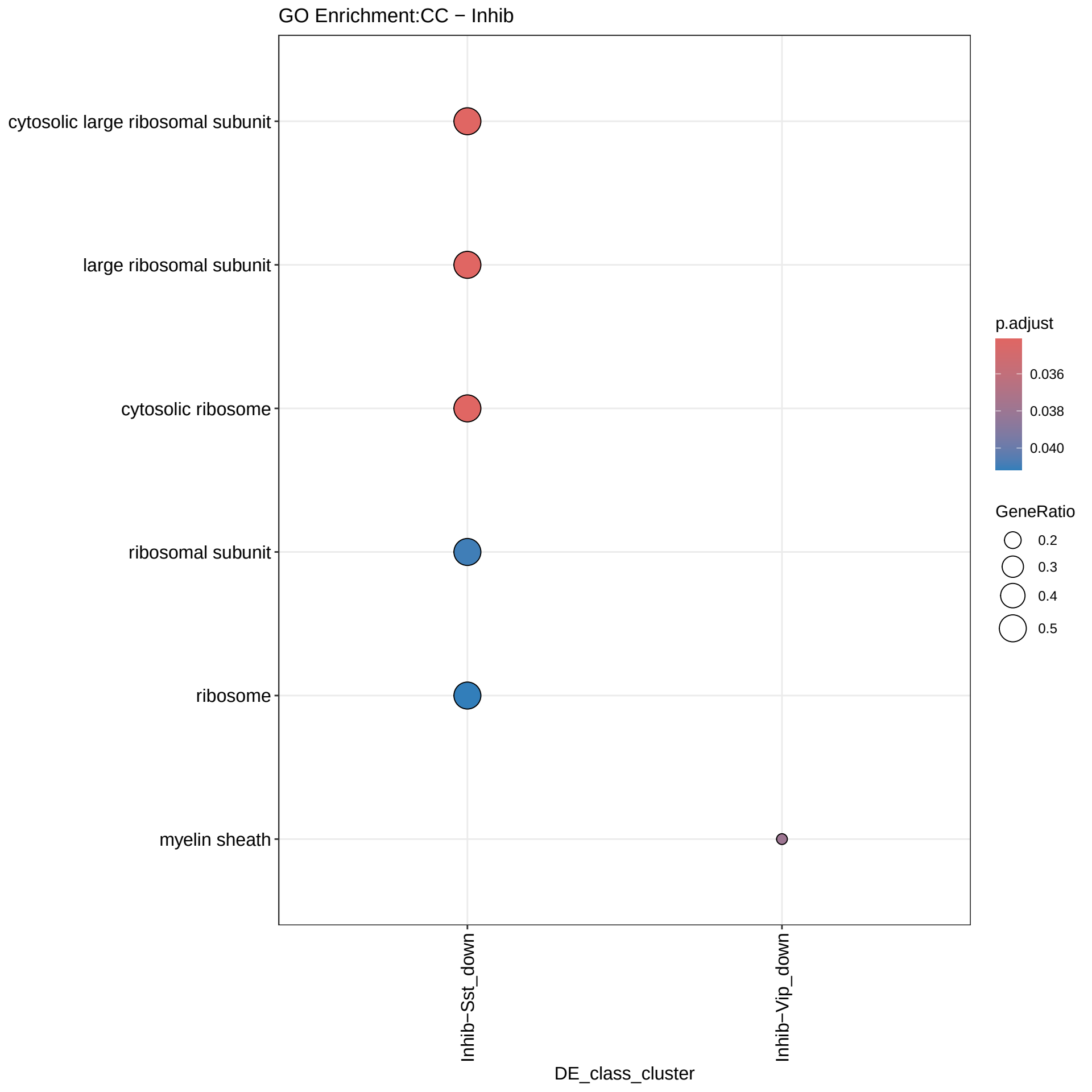

GO Enrichment:BP − Vasc
regulation of protein localization to early endosome
positive regulation of protein localization to early endosome
protein localization to early endosome
regulation of receptor catabolic process
positive regulation of protein localization to endosome
skin morphogenesis
protein trimerization
protein heterooligomerization
extracellular matrix assembly
collagen fibril organization
steroid catabolic process
inorganic anion transport
lung development
respiratory tube development
respiratory system development
negative regulation of epidermal cell differentiation
negative regulation of epidermis development
metanephric tubule morphogenesis
regulation of timing of cell differentiation
positive regulation of astrocyte differentiation
Vasc−Endo_down
Vasc−Endo_up
Vasc−VLMC_down
Vasc−VLMC_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
0.04
GeneRatio
0.5
0.6
0.7
0.8
0.9
1.0

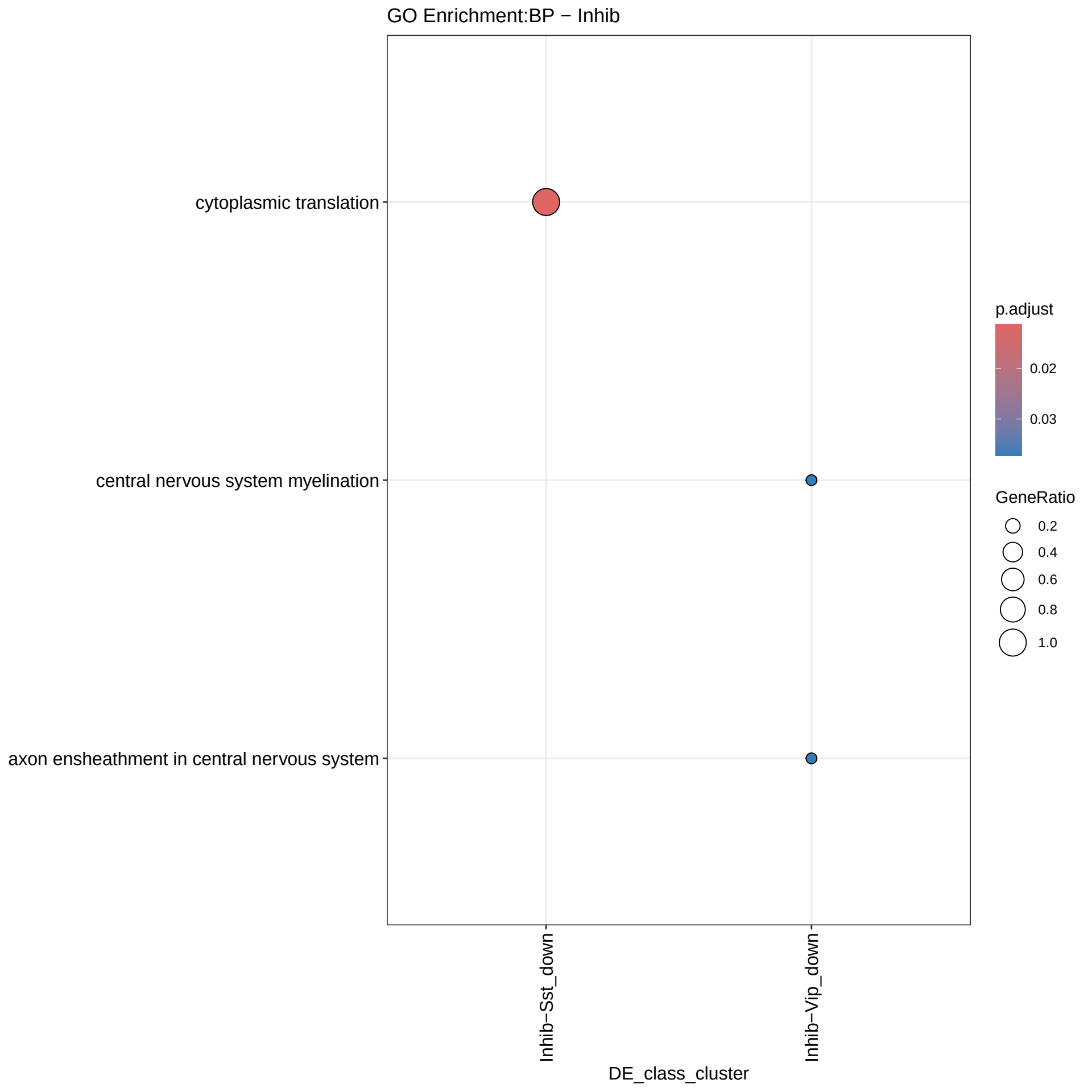

GO Enrichment:MF − Vasc
histone ubiquitin ligase activity
enzyme−substrate adaptor activity
histone binding
ubiquitin protein ligase activity
ubiquitin−like protein ligase binding
platelet−derived growth factor binding
extracellular matrix structural constituent conferring tensile strength
SMAD binding
protease binding
growth factor binding
phosphate transmembrane transporter activity
steroid dehydrogenase activity, acting on the CH−OH group of donors, NAD or NADP as acceptor
steroid dehydrogenase activity
NADP binding
metalloendopeptidase activity
JUN kinase binding
E−box binding
transcription corepressor binding
transcription coregulator binding
histone deacetylase binding
Vasc−Endo_down
Vasc−Endo_up
Vasc−VLMC_down
Vasc−VLMC_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.5
0.6
0.7
0.8
0.9
1.0


GO Enrichment:CC − Excit
dendritic spine
neuron spine
AMPA glutamate receptor complex
ionotropic glutamate receptor complex
neurotransmitter receptor complex
plasma membrane signaling receptor complex
monoatomic ion channel complex
transmembrane transporter complex
transporter complex
respiratory chain complex
oxidoreductase complex
mitochondrial inner membrane
vacuolar proton−transporting V−type ATPase, V1 domain
proton−transporting V−type ATPase, V1 domain
proton−transporting two−sector ATPase complex, catalytic domain
extrinsic component of synaptic vesicle membrane
Golgi cis cisterna
Excit−L2_5−1_down
Excit−L2_5−2_up
Excit−L5−2_down
Excit−L5−2_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.4
0.5
0.6
0.7
0.8
0.9
1.0


GO Enrichment:BP − Excit
negative regulation of BMP signaling pathway
positive regulation of neuron differentiation
negative regulation of cellular response to growth factor stimulus
regulation of BMP signaling pathway
BMP signaling pathway
oxidative phosphorylation
proton transmembrane transport
aerobic electron transport chain
aerobic respiration
ATP synthesis coupled electron transport
endosomal lumen acidification
Golgi lumen acidification
synaptic vesicle lumen acidification
lysosomal lumen acidification
synaptic vesicle maturation
Excit−L2_5−2_up
Excit−L5−2_down
Excit−L5−2_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.5
0.6
0.7
0.8
0.9
1.0


GO Enrichment:MF − Excit
transmembrane receptor protein tyrosine kinase adaptor activity
bioactive lipid receptor activity
G protein−coupled receptor activity
CoA−ligase activity
phospholipid binding
oxidoreduction−driven active transmembrane transporter activity
proton transmembrane transporter activity
electron transfer activity
active monoatomic ion transmembrane transporter activity
primary active transmembrane transporter activity
ATPase−coupled ion transmembrane transporter activity
ATPase activity, coupled to transmembrane movement of ions, rotational mechanism
proton−transporting ATPase activity, rotational mechanism
ATPase−coupled monoatomic cation transmembrane transporter activity
ATPase−coupled transmembrane transporter activity
Excit−L2_5−1_down
Excit−L5−2_down
Excit−L5−2_up
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.2
0.4
0.6
0.8
1.0


GO Enrichment:CC − Inhib
cytosolic large ribosomal subunit
large ribosomal subunit
cytosolic ribosome
ribosomal subunit
ribosome
myelin sheath
Inhib−Sst_down
Inhib−Vip_down
DE_class_cluster
p.adjust
0.036
0.038
0.040
GeneRatio
0.2
0.3
0.4
0.5


GO Enrichment:BP − Inhib
cytoplasmic translation
central nervous system myelination
axon ensheathment in central nervous system
p.adjust
0.02
0.03
GeneRatio
0.2
0.4
0.6
0.8
1.0
Inhib−Sst_down
Inhib−Vip_down
DE_class_cluster